material to form an **F′ plasmid** (**figure 13.15*a***). It is not unusual to observe the inclusion of one or more genes in excised F plasmids. The F′ cell retains all of its genes, although some of them are on the plasmid, and still mates only with an F^- recipient. F′ $\times$ F^- conjugation is virtually identical with $F^+ \times F^-$ mating. Once again, the plasmid is transferred, but usually bacterial genes on the chromosome are not (figure 13.15*b*). Bacterial genes on the F′ plasmid are transferred with it and need not be incorporated into the recipient chromosome to be expressed. The recipient becomes F′ and is a partially diploid merozygote since it has two sets of the genes carried by the plasmid. In this way specific bacterial genes may spread rapidly throughout a bacterial population. Such transfer of bacterial genes is often called sexduction.

F′ conjugation is very important to the microbial geneticist. A partial diploid's behavior shows whether the allele carried by an F′ plasmid is dominant or recessive to the chromosomal gene. The formation of F′ plasmids also is useful in mapping the chromosome since if two genes are picked up by an F factor they must be neighbors.

1. What is bacterial conjugation and how was it discovered?
2. Distinguish between F^+, Hfr, and F^- strains of *E. coli* with respect to their physical nature and role in conjugation.
3. Describe in some detail how $F^+ \times F^-$ and Hfr conjugation processes proceed, and distinguish between the two in terms of mechanism and the final results.
4. What is F′ conjugation and why is it so useful to the microbial geneticist? How does the F′ plasmid differ from a regular F plasmid? What is sexduction?

13.5 DNA Transformation

The second way in which DNA can move between bacteria is through transformation, discovered by Fred Griffith in 1928. **Transformation** is the uptake by a cell of a naked DNA molecule or fragment from the medium and the incorporation of this molecule into the recipient chromosome in a heritable form. In natural transformation the DNA comes from a donor bacterium. The process is random, and any portion of a genome may be transferred between bacteria. The discovery of transformation (pp. 228–29)

When bacteria lyse, they release considerable amounts of DNA into the surrounding environment. These fragments may be relatively large and contain several genes. If a fragment contacts a **competent** cell, one able to take up DNA and be transformed, it can be bound to the cell and taken inside (**figure 13.16*a***). The transformation frequency of very competent cells is around 10^{-3} for most genera when an excess of DNA is used. That is, about one cell in every thousand will take up and integrate the gene. Competency is a complex phenomenon and is dependent on several conditions. Bacteria need to be in a certain stage of growth; for example, *S. pneumoniae* becomes competent during the exponential phase when the population reaches about 10^7 to 10^8 cells per ml. When a population becomes competent, bacteria such as pneumococci secrete a small protein called the competence factor that stimulates the production

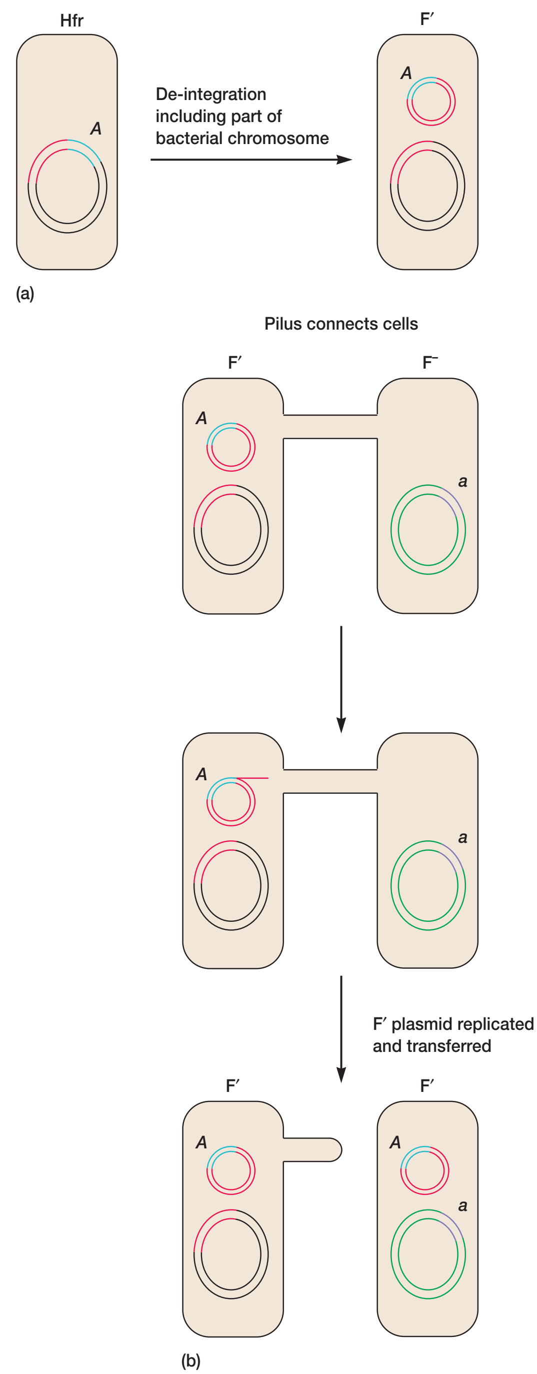


Figure 13.15 F′ Conjugation. (**a**) Due to an error in excision, the *A* gene of an Hfr cell is picked up by the F factor. (**b**) The *A* gene is then transferred to a recipient during conjugation.

of 8 to 10 new proteins required for transformation. Natural transformation has been discovered so far only in certain gram-positive and gram-negative genera: *Streptococcus, Bacillus, Thermoactinomyces, Haemophilus, Neisseria, Moraxella, Acinetobacter,*

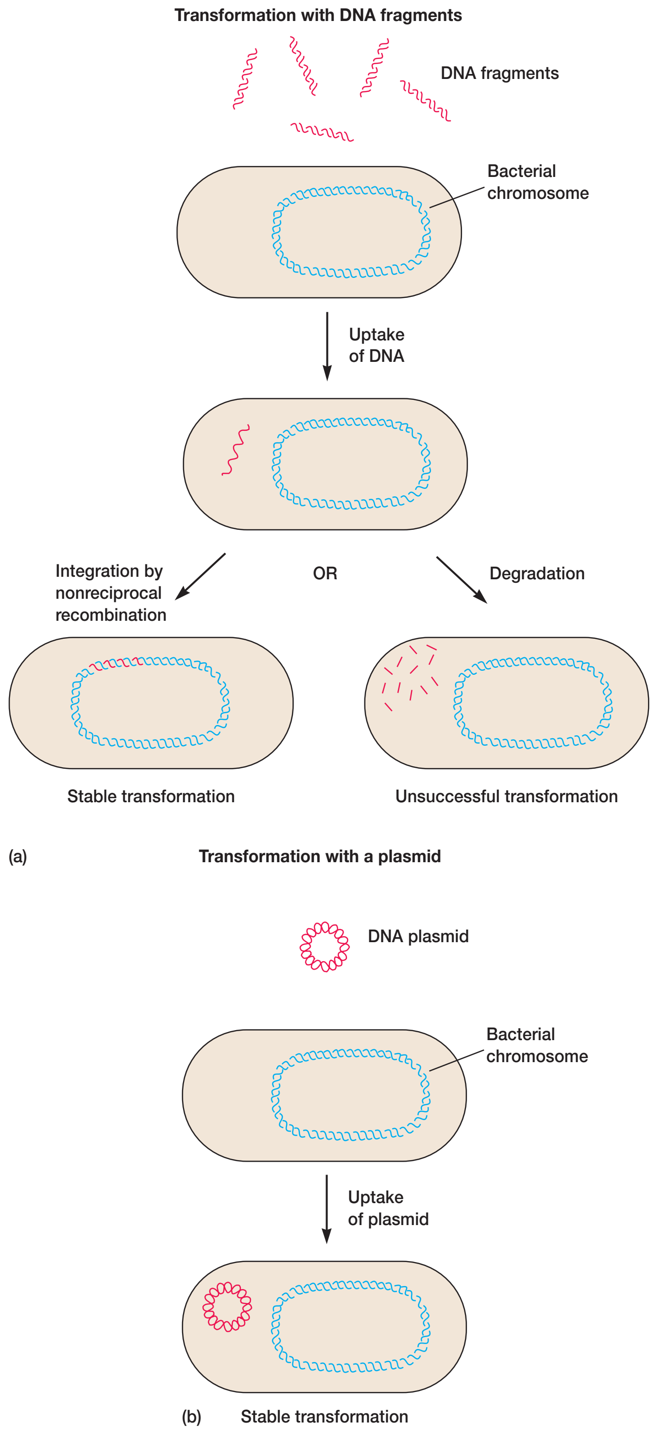


Figure 13.16 Bacterial Transformation. Transformation with (**a**) DNA fragments and (**b**) plasmids. Transformation with a plasmid often is induced artificially in the laboratory. The transforming DNA is in red and integration is at a homologous region of the genome. See text for details.

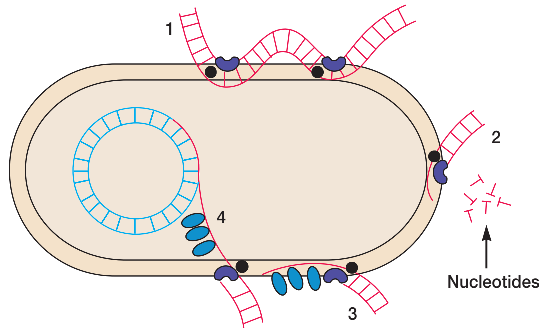


Figure 13.17 The Mechanism of Transformation. (*1*) A long double-stranded DNA molecule binds to the surface with the aid of a DNA-binding protein (●) and is nicked by a nuclease ( ). (*2*) One strand is degraded by the nuclease. (*3*) The undegraded strand associates with a competence-specific protein ( ). (*4*) The single strand enters the cell and is integrated into the host chromosome in place of the homologous region of the host DNA.

Azotobacter, and *Pseudomonas.* Other genera also may be capable of transformation. Gene transfer by this process occurs in soil and marine environments and may be an important route of genetic exchange in nature.

The mechanism of transformation has been intensively studied in *S. pneumoniae* (**figure 13.17**). A competent cell binds a double-stranded DNA fragment if the fragment is moderately large; the process is random, and donor fragments compete with each other. The DNA then is cleaved by endonucleases to double-stranded fragments about 5 to 15 kilobases in size. DNA uptake requires energy expenditure. One strand is hydrolyzed by an envelope-associated exonuclease during uptake; the other strand associates with small proteins and moves through the plasma membrane. The single-stranded fragment can then align with a homologous region of the genome and be integrated, probably by a mechanism similar to that depicted in figure 13.3.

Transformation in *Haemophilus influenzae,* a gram-negative bacterium, differs from that in *S. pneumoniae* in several respects. *Haemophilus* does not produce a competence factor to stimulate the development of competence, and it takes up DNA from only closely related species (*S. pneumoniae* is less particular about the source of its DNA). Double-stranded DNA, complexed with proteins, is taken in by membrane vesicles. The specificity of *Haemophilus* transformation is due to a special 11 base pair sequence (5′AAGTGCGGTCA3′) that is repeated over 1,400 times in *H. influenzae* DNA. DNA must have this sequence to be bound by a competent cell.

Artificial transformation is carried out in the laboratory by a variety of techniques, including treatment of the cells with calcium chloride, which renders their membranes more permeable to DNA. This approach succeeds even with species that are not naturally competent, such as *E. coli.* Relatively high concentrations of DNA, higher than would normally be present in nature, are used to increase transformation frequency. When linear DNA fragments are to be used in transformation, *E. coli* usually is ren-

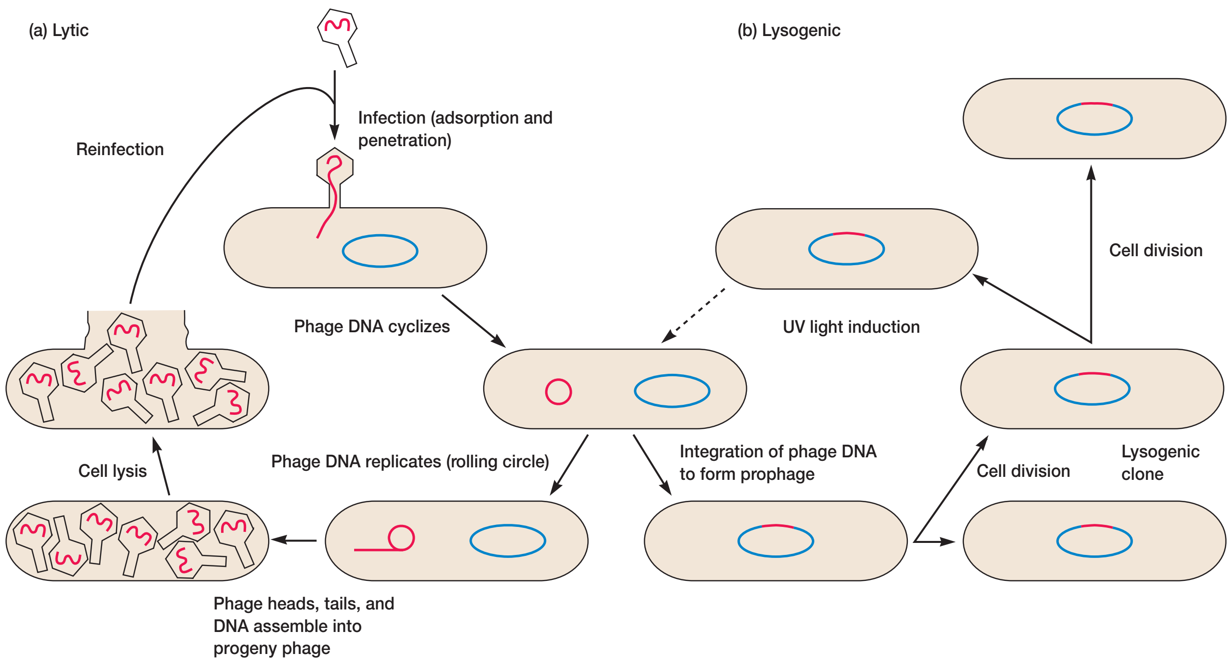


Figure 13.18 Lytic versus Lysogenic Infection by Phage Lambda. (**a**) Lytic infection. (**b**) Lysogenic infection. Viral and prophage DNA are in red.

dered deficient in one or more exonuclease activities to protect the transforming fragments. It is even easier to transform bacteria with plasmid DNA since plasmids are not as easily degraded as linear fragments and can replicate within the host (figure 13.16*b*). This is a common method for introducing recombinant DNA into bacterial cells (*see sections 14.5 and 14.7*). DNA from any source can be introduced into bacteria by splicing it into a plasmid before transformation.

1. Define transformation and competence.
2. Describe how transformation occurs in *S. pneumoniae.* How does the process differ in *H. influenzae?*
3. Discuss two ways in which artificial transformation can be used to place functional genes within bacterial cells.

13.6 Transduction

Bacterial viruses or bacteriophages participate in the third mode of bacterial gene transfer. These viruses have relatively simple structures in which virus genetic material is enclosed within an outer coat, composed mainly or solely of protein. The coat protects the genome and transmits it between host cells. The morphology and life cycle of bacteriophages is not discussed in detail until chapter 17. Nevertheless, it is necessary to briefly describe the life cycle here as background for a consideration of the bacteriophage's role in gene transfer. The lytic cycle (pp. 382–88); Lysogeny (pp. 390–95)

After infecting the host cell, a bacteriophage (phage for short) often takes control and forces the host to make many copies of the virus. Eventually the host bacterium bursts or lyses and releases new phages. This reproductive cycle is called a lytic cycle because it ends in lysis of the host. The cycle has four phases (**figure 13.18*a***). First, the virus particle attaches to a specific receptor site on the bacterial surface. The genetic material, which is often double-stranded DNA, then enters the cell. After adsorption and penetration, the virus chromosome forces the bacterium to make virus nucleic acids and proteins. The third stage begins after the synthesis of virus components. Phages are assembled from these components. The assembly process may be complex, but in all cases phage nucleic acid is packed within the virus's protein coat. Finally, the mature viruses are released by cell lysis.

Bacterial viruses that reproduce using a lytic cycle often are called virulent bacteriophages because they destroy the host cell. Many DNA phages, such as the lambda phage (*see p. 391*), are also capable of a different relationship with their host (figure 13.18*b*). After adsorption and penetration, the viral genome does not take control of its host and destroy it while producing new phages. Instead the genome remains within the host cell and is reproduced along with the bacterial chromosome. A clone of infected cells arises and may grow for long periods while appearing perfectly normal. Each of these infected bacteria can produce phages and lyse under appropriate environmental conditions. This relationship between the phage and its host is called **lysogeny.**

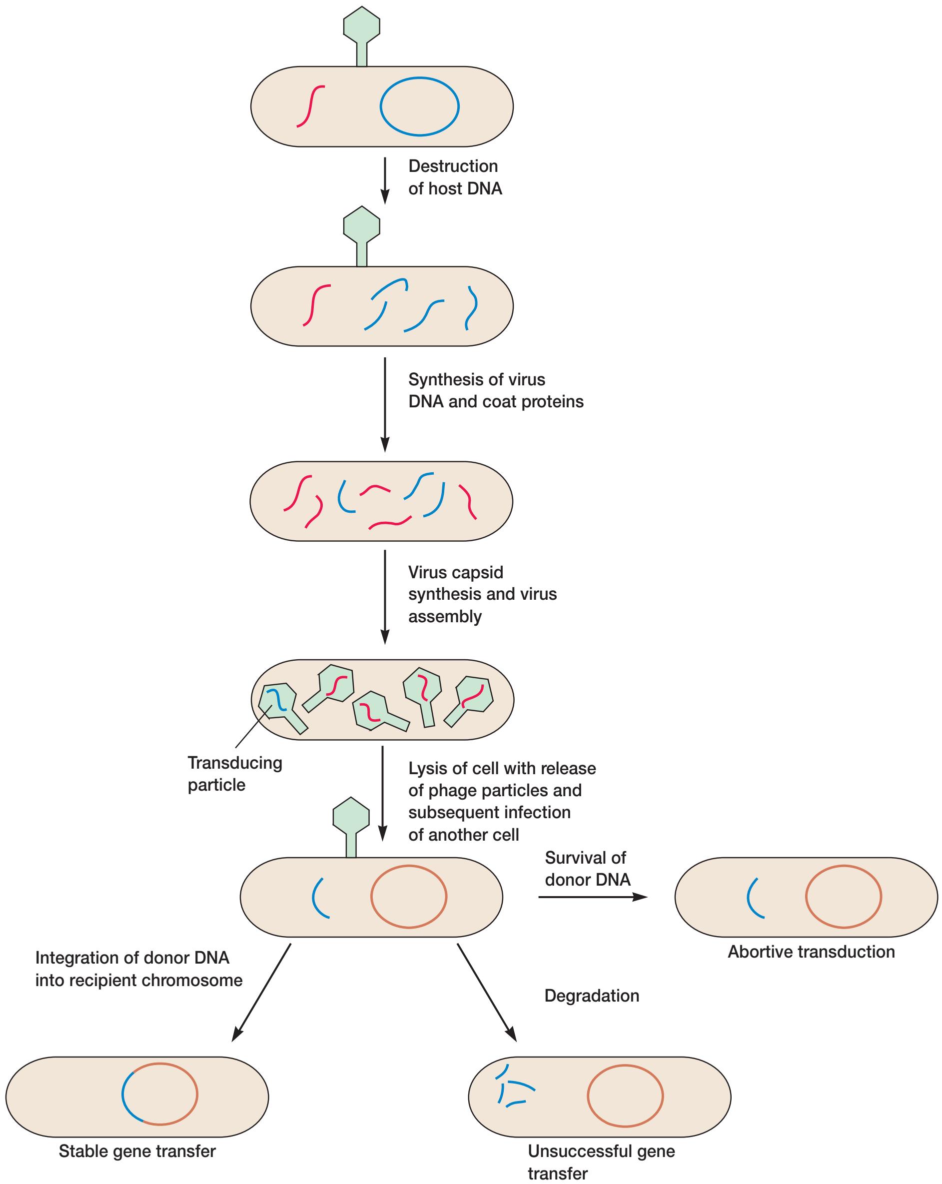


Figure 13.19 **Generalized Transduction by Bacteriophages.** See text for details.

Bacteria that can produce phage particles under some conditions are said to be **lysogens** or **lysogenic,** and phages able to establish this relationship are **temperate phages.** The latent form of the virus genome that remains within the host without destroying it is called the **prophage.** The prophage usually is integrated into the bacterial genome (figure 13.18*b*). Sometimes phage reproduction is triggered in a lysogenized culture by exposure to UV radiation or other factors. The lysogens are then destroyed and new phages released. This phenomenon is called induction.

Transduction is the transfer of bacterial genes by viruses. Bacterial genes are incorporated into a phage capsid because of errors made during the virus life cycle. The virus containing these genes then injects them into another bacterium, completing the transfer. Transduction may be the most common mechanism for gene exchange and recombination in bacteria. There are two very different kinds of transduction: generalized and specialized.

Generalized Transduction

Generalized transduction (**figure 13.19**) occurs during the lytic cycle of virulent and temperate phages and can transfer any part of the bacterial genome. During the assembly stage, when the viral chromosomes are packaged into protein capsids, random fragments of the partially degraded bacterial chromosome also may be packaged by mistake. Because the capsid can contain only a limited quantity of DNA, the viral DNA is left behind. The quantity of bacterial DNA carried depends primarily on the size of the capsid. The P22 phage of *Salmonella typhimurium* usually carries about 1% of the bacterial genome; the P1 phage of *E. coli* and a variety of gram-negative bacteria carries about 2.0 to 2.5% of the genome. The resulting virus particle often injects the DNA into another bacterial cell but does not initiate a lytic cycle. This phage is known as a **generalized transducing particle** or phage and is simply a carrier of genetic information from the original bacterium to another cell. As in transformation, once the DNA has been injected, it must be incorporated into the recipient cell's chromosome to preserve the transferred genes. The DNA remains double stranded during transfer, and both strands are integrated into the endogenote's genome. About 70 to 90% of the transferred DNA is not integrated but often is able to survive and express itself. **Abortive transductants** are bacteria that contain this nonintegrated, transduced DNA and are partial diploids.

Generalized transduction was discovered in 1951 by Joshua Lederberg and Norton Zinder during an attempt to show that conjugation, discovered several years earlier in *E. coli,* could occur in other bacterial species. Lederberg and Zinder were repeating the earlier experiments with *Salmonella typhimurium.* They found that incubation of a mixture of two multiply auxotrophic strains yielded prototrophs at the level of about one in 10^5. This seemed like good evidence for bacterial recombination, and indeed it was, but their initial conclusion that the transfer resulted from conjugation was not borne out. When these investigators performed the U-tube experiment (figure 13.13) with *Salmonella,* they still recovered prototrophs. The filter in the U tube had small enough pores to block the movement of bacteria between the two sides but allowed the phage P22 to pass. Lederberg and Zinder had intended to confirm that conjugation was present in another bacterial species and had instead discovered a completely new mechanism of bacterial gene transfer. The seemingly routine piece of research led to surprising and important results. A scientist must always keep an open mind about results and be prepared for the unexpected.

Specialized Transduction

In **specialized** or **restricted transduction,** the transducing particle carries only specific portions of the bacterial genome. Specialized transduction is made possible by an error in the lysogenic life cycle. When a prophage is induced to leave the host chromosome, excision is sometimes carried out improperly. The resulting phage genome contains portions of the bacterial chromosome (about 5 to 10% of the bacterial DNA) next to the integration site, much like the situation with F′ plasmids (**figure 13.20**). A transducing phage genome usually is defective and lacks some part of its attachment site. The transducing particle will inject bacterial genes into another bacterium, even though the defective phage cannot reproduce without assistance. The bacterial genes may become stably incorporated under the proper circumstances.

The best-studied example of specialized transduction is the lambda phage. The lambda genome inserts into the host chromosome at specific locations known as attachment or *att* sites (**figure 13.21;** *see also figures 17.16 and 17.20*). The phage *att* sites and bacterial *att* sites are similar and can complex with each other, although they are not identical. The *att* site for lambda is next to the *gal* and *bio* genes on the *E. coli* chromosome; consequently, specialized transducing lambda phages most often carry these bacterial genes. The lysate, or product of cell lysis, resulting from the induction of lysogenized *E. coli* contains normal phage and a few defective transducing particles. These particles are called lambda *dgal* because they carry the galactose utilization genes (figure 13.21). Because these lysates contain only a few transducing particles, they often are called **low-frequency transduction lysates (LFT lysates).** Whereas the normal phage has a complete *att* site, defective transducing particles have a nonfunctional hybrid integration site that is part bacterial and part phage in origin. Integration of the defective phage chromosome does not readily take place. Transducing phages also may have lost some genes essential for reproduction. Stable transductants can arise from recombination between the phage and the bacterial chromosome because of crossovers on both sides of the *gal* site (figure 13.21). The lysogenic cycle of lambda phage (pp. 391–95)

Defective lambda phages carrying the *gal* gene can integrate if there is a normal lambda phage in the same cell. The normal phage will integrate, yielding two bacterial/phage hybrid *att* sites where the defective lambda *dgal* phage can insert (figure 13.21). It also supplies the genes missing in the defective phage. The normal phage in this instance is termed the **helper phage** because it aids integration and reproduction of the defective phage. These transductants are unstable because the prophages can be induced to excise by agents such as UV radiation. Excision, however, produces a lysate containing a fairly equal mixture of defective lambda *dgal* phage and normal helper phage. Because it is very effective in transduction, the lysate is called a **high-frequency transduction lysate (HFT lysate).** Reinfection of bacteria with this mixture will result in the generation of considerably more transductants. LFT lysates and those produced by generalized transduction have one transducing particle in 10^5 or 10^6 phages; HFT lysates contain transducing particles with a frequency of about 0.1 to 0.5.

1. Briefly describe the lytic and lysogenic viral reproductive cycles. Define lysogeny, lysogen, temperate phage, prophage, and transduction.
2. Describe generalized transduction, how it occurs, and the way in which it was discovered. What is an abortive transductant?
3. What is specialized or restricted transduction and how does it come about? Be able to distinguish between LFT and HFT lysates and describe how they are formed.

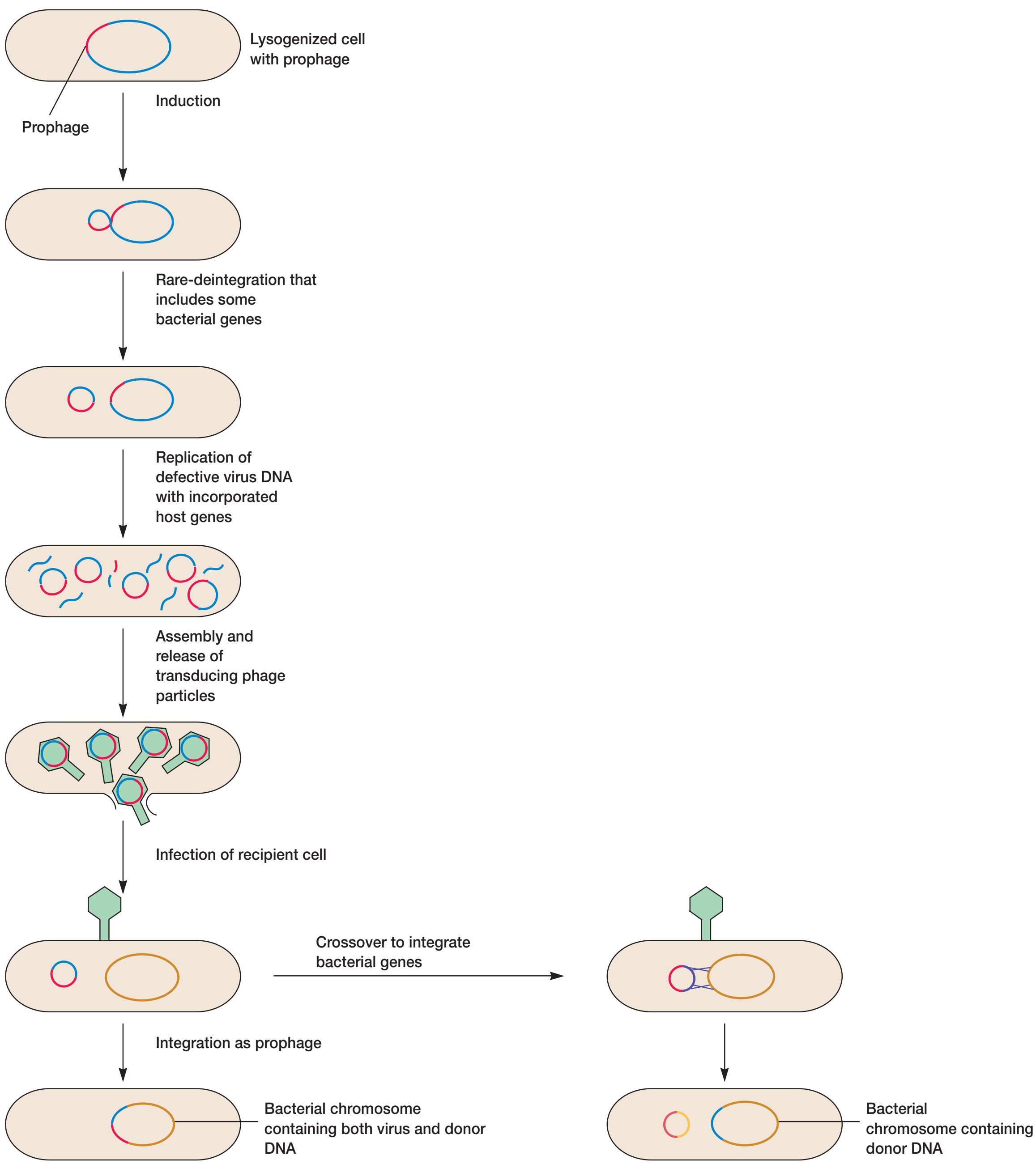


Figure 13.20 Specialized Transduction by a Temperate Bacteriophage. Recombination can produce two types of transductants. See text for details.

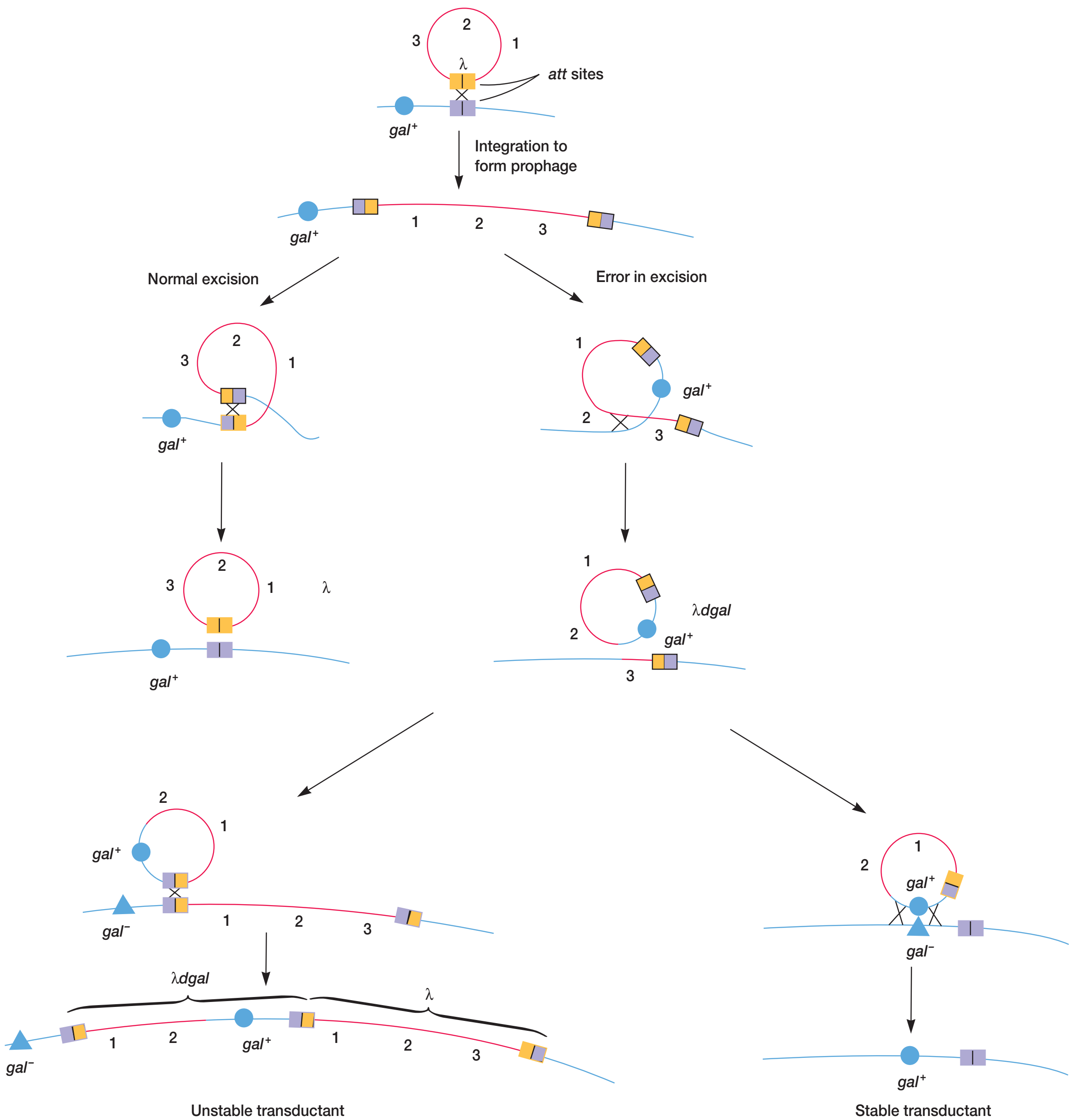
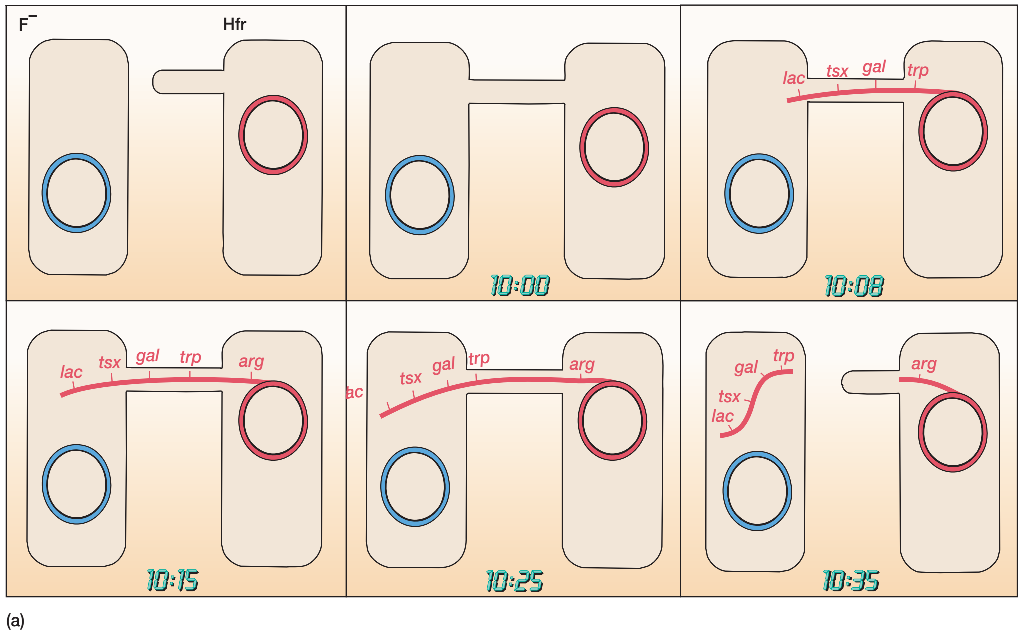


Figure 13.21 The Mechanism of Transduction for Phage Lambda and *E. coli*. Integrated lambda phage lies next to the *gal* genes. When it excises normally (top left), the new phage is complete and contains no bacterial genes. Rarely excision occurs asymmetrically (top right), and the *gal* genes are then picked up and some phage genes are lost. The result is a defective lambda phage that carries bacterial genes and can transfer them to a new recipient. See text for details.



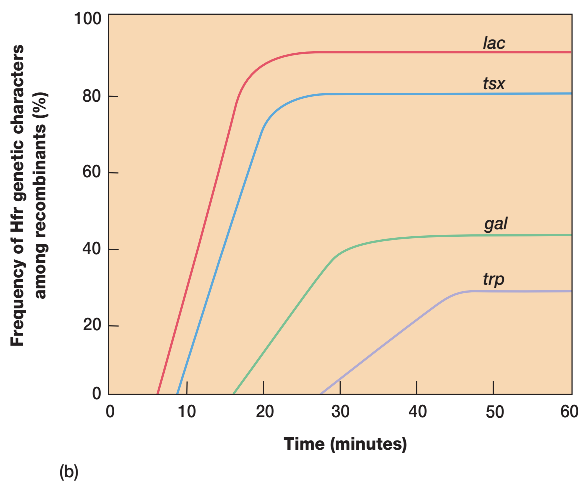


Figure 13.22 The Interrupted Mating Experiment. An interrupted mating experiment on Hfr × F⁻ conjugation. (**a**) The linear transfer of genes is stopped by breaking the conjugation bridge to study the sequence of gene entry into the recipient cell. (**b**) An example of the results obtained by an interrupted mating experiment. The gene order is *lac-tsx-gal-trp*.

13.7 Mapping the Genome

Finding the location of genes in any organism's genome is a very complex task. This section surveys approaches to mapping the bacterial genome, using *E. coli* as an example. All three modes of gene transfer and recombination have been used in mapping. Gene structure and the nature of mutations (pp. 241–53)

Hfr conjugation is frequently used to map the relative location of bacterial genes. This technique rests on the observation that during conjugation the linear chromosome moves from donor to recipient at a constant rate. In an **interrupted mating experiment** the conjugation bridge is broken and Hfr × F⁻ mating is stopped at various intervals after the start of conjugation by mixing the culture vigorously in a blender (**figure 13.22*a***). The order and timing of gene transfer can be determined because they are a direct reflection of the order of genes on the bacterial chromosome (figure 13.22*b*). For example, extrapolation of the curves in figure 13.22*b* back to the x-axis will give the time at which each gene just began to enter the recipient. The result is a circular chromosome map with distances expressed in terms of the minutes elapsed until a gene is transferred. This technique can fairly precisely locate genes 3 minutes or more apart. The heights of the plateaus in figure 13.22*b* are lower for genes that are more distant from the F factor (the origin of transfer) because there is an ever-greater chance that the conjugation bridge will spontaneously break as transfer continues. Because of the relatively large size of the *E. coli* genome, it is not possible to generate a map from one Hfr strain. Therefore several Hfr strains with the F plasmid integrated at different locations must be used and their maps superimposed on one another. The overall map is adjusted to 100 minutes, although complete transfer may require somewhat more than 100 minutes. In a sense, minutes are an indication of map distance and not strictly a measure of time. Zero time is set at the threonine (*thr*) locus.

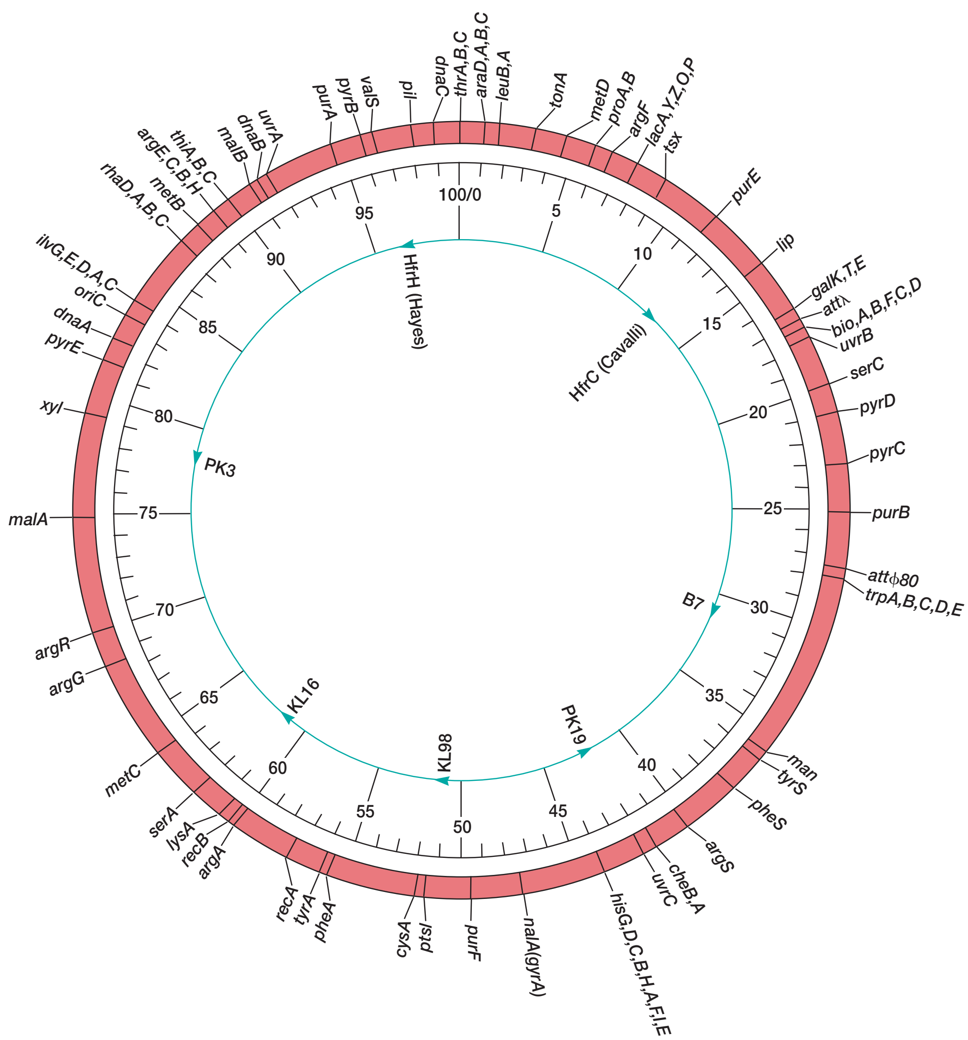


Figure 13.23 ***E. coli*** **Genetic Map.** A circular genetic map of *E. coli* K12 with the location of selected genes. The inner circle shows the origin and direction of transfer of several Hfr strains. The map is divided into 100 minutes, the time required to transfer the chromosome from an Hfr cell to F^- at 37°C.

Gene linkage, or the proximity of two genes on a chromosome, also can be determined from transformation by measuring the frequency with which two or more genes simultaneously transform a recipient cell. Consider the case for cotransformation by two genes. In theory, a bacterium could simultaneously receive two genes, each carried on a separate DNA fragment. However, it is much more likely that the genes reside on the same fragment. If two genes are closely linked on the chromosome, then they should be able to cotransform. The closer the genes are together, the more often they will be carried on the same fragment and the higher will be the frequency of cotransformation. If genes are spaced a great distance apart, they will be carried on separate DNA fragments and the frequency of double transformants will equal the product of the individual transformation frequencies.

Generalized transduction can be used to obtain linkage information in much the same way as transformation. Linkages usually are expressed as cotransduction frequencies, using the argument that the closer two genes are to each other, the more likely they both will reside on the DNA fragment incorporated into a single phage capsid. The *E. coli* phage P1 is often used in such mapping because it can randomly transduce up to 1 to 2% of the genome.

Specialized transduction is used to find which phage attachment site is close to a specific gene. The relative locations of specific phage *att* sites are known from conjugational mapping, and the genes linked to each *att* site can be determined by means of specialized transduction. These data allow precise placement of genes on the chromosome.

A simplified genetic map of *E. coli* K12 is given in **figure 13.23.** Because conjugation data are not high resolution and cannot be used to position genes that are very close together, the whole map is developed using several mapping techniques. Usually, interrupted mating data are combined with those from cotransduction and cotransformation studies. Data from recombination studies also are used. Normally a new genetic marker in the *E. coli* genome is located within a relatively small region of the genome (10 to 15 minutes long) using a series of Hfr strains with F factor integration sites scattered throughout the genome. Once the genetic marker has been located with respect to several genes in the same region, its position relative to nearby neighbors is more accurately determined using transformation and transduction studies. Recent maps of the *E. coli* chromosome give the locations of more than a thousand genes. Remember that the genetic map only depicts physical reality in a relative sense. A map unit in one region of the genome may not be the same physical distance as a unit in another part.

Genetic maps provide useful information in addition to the order of the genes. For example, there is considerable clustering of genes in *E. coli* K12 (figure 13.23). In the regions around 2, 17, and 27 minutes, there are many genes, whereas relatively few genetic markers are found in the 33 minute region. The areas apparently lacking genes may well have undiscovered genes, but perhaps their function is not primarily that of coding genetic information. One hypothesis is that the 33 minute region is involved in attachment of the *E. coli* chromosome to the plasma membrane during replication and cell division. It is interesting that this region is almost exactly opposite the origin of replication for the chromosome (*oriC*).

Of course a great deal could be learned by comparing a microorganism's genetic map with the actual nucleotide sequence of its genome. It is now possible to fairly rapidly sequence procaryotic genomes; genome sequencing and analysis will be discussed later in some detail. Microbial genomics (chapter 15)

13.8 Recombination and Genome Mapping in Viruses

Bacteriophage genomes also undergo recombination, although the process is different from that in bacteria. Because phages themselves reproduce within cells and cannot recombine directly, crossing-over must occur inside a host cell. In principle, a virus recombination experiment is easy to carry out. If bacteria are mixed with enough phages, at least two virions will infect each cell on the average and genetic recombination should be observed. Phage progeny in the resulting lysate can be checked for alternate combinations of the initial parental genotypes. Bacteriophage lytic cycle (pp. 382–88)

Alfred Hershey initially demonstrated recombination in the phage T2, using two strains with differing phenotypes. Two of the parental strains in Hershey's crosses were h^+r^+ and *hr* (**figure 13.24*a***). The gene *h* influences host range; when gene *h* changes, T2 infects different strains of *E. coli.* Phages with the r^+ gene have wild type plaque morphology, whereas T2 with the *r* genotype has a rapid lysis phenotype and produces larger than normal plaques with sharp edges (figures 13.24*b* and 13.24*c*). In one experiment Hershey infected *E. coli* with large quantities of the h^+r^+ and *hr* T2 strains (figure 13.24*a*). He then plated out the lysates with a mixture of two different host strains and was able to detect significant numbers of h^+r and hr^+ recombinants, as well as parental type plaques. As long as there are detectable phenotypes and methods for carrying out the crosses, it is possible to map phage genes in this way. Plaque formation and morphology (p. 364)

Phage genomes are so small that often it is convenient to map them without determining recombination frequencies. Some techniques actually generate physical maps, which often are most useful in genetic engineering. Several of these methods require manipulation of the DNA with subsequent examination in the electron microscope. For example, one can directly compare wild-type and mutant viral chromosomes. In heteroduplex mapping the two types of chromosomes are denatured, mixed, and allowed to rejoin or anneal. When joined, the homologous regions of the different DNA molecules form a regular double helix. In locations where the bases do not pair due to the presence of a mutation such as a deletion or insertion, bubbles will be visible in the electron microscope.

Several other direct techniques are used to map viral genomes or parts of them. Restriction endonucleases (*see section 14.1*) are employed together with electrophoresis to analyze DNA fragments and locate deletions and other mutations that affect electrophoretic mobility. Phage genomes also can be directly sequenced to locate particular mutations and analyze the changes that have taken place.

It should be noted that many of these physical mapping techniques also have been employed in the analysis of relatively small portions of bacterial genomes. Furthermore, these methods are useful to the genetic engineer who is concerned with direct manipulation of DNA.

1. Describe how the bacterial genome can be mapped using Hfr conjugation, transformation, generalized transduction, and specialized transduction. Include both a description of each technique and any assumptions underlying its use.
2. Why is it necessary to use several different techniques in genome mapping? How is this done in practice?
3. How does recombination in viruses differ from that in bacteria? How did Hershey first demonstrate virus recombination?
4. Describe heteroduplex mapping.

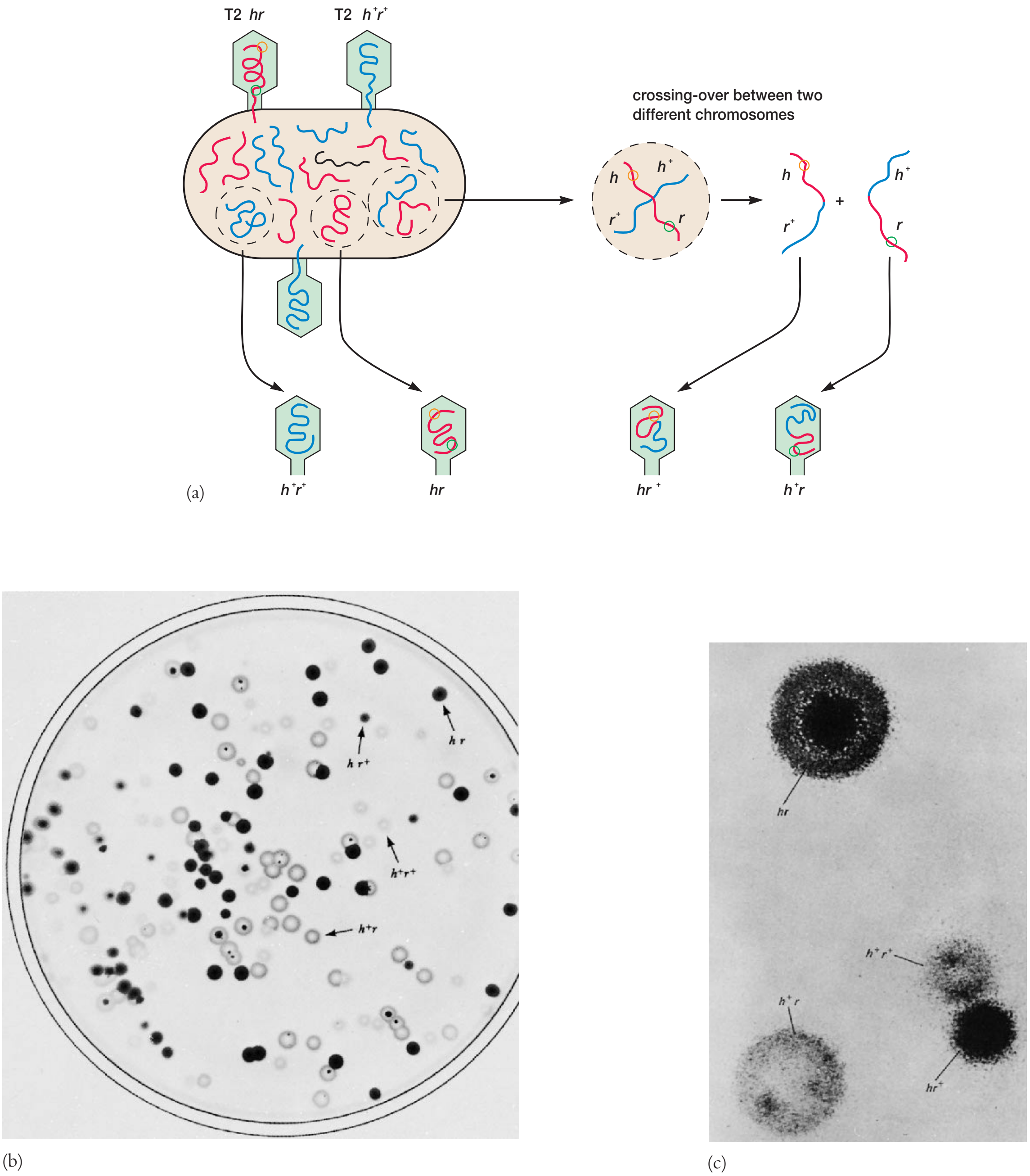


Figure 13.24 Genetic Recombination in Bacteriophages. (**a**) A summary of a genetic recombination experiment with the *hr* and h^+r^+ strains of the T2 phage. The *hr* chromosome is shown in color. (**b**) The types of plaques produced by this experiment on a lawn of *E. coli.* (**c**) A close-up of the four plaque types. See text for details.

Summary

1. In recombination, genetic material from two different chromosomes is combined to form a new, hybrid chromosome. There are three types of recombination: general recombination, site-specific recombination, and replicative recombination.
2. Bacterial recombination is a one-way process in which the exogenote is transferred from the donor to a recipient and integrated into the endogenote (**figure 13.4**).
3. Plasmids are small, circular, autonomously replicating DNA molecules that can exist independent of the host chromosome. Their genes are not required for host survival.
4. Episomes are plasmids that can be reversibly integrated with the host chromosome.
5. Many important types of plasmids have been discovered: F factors, R factors, Col plasmids, virulence plasmids, and metabolic plasmids.
6. Transposons or transposable elements are DNA segments that move about the genome in a process known as transposition.
7. There are two types of transposable elements: insertion sequences and composite transposons.
8. Transposable elements cause mutations, block translation and transcription, turn genes on and off, aid F plasmid insertion, and carry antibiotic resistance genes.
9. Conjugation is the transfer of genes between bacteria that depends upon direct cell-cell contact mediated by the F pilus.
10. In $F^+ \times F^-$ mating the F factor remains independent of the chromosome and a copy is transferred to the F^- recipient; donor genes are not usually transferred (**figure 13.14**).
11. Hfr strains transfer bacterial genes to recipients because the F factor is integrated into the host chromosome. A complete copy of the F factor is not often transferred (**figure 13.14**).
12. When the F factor leaves an Hfr chromosome, it occasionally picks up some bacterial genes to become an F′ plasmid, which readily transfers these genes to other bacteria (**figure 13.15**).
13. Transformation is the uptake of a naked DNA molecule by a competent cell and its incorporation into the genome (**figure 13.16**).
14. Bacterial viruses or bacteriophages can reproduce and destroy the host cell (lytic cycle) or become a latent prophage that remains within the host (lysogenic cycle) (**figure 13.18**).
15. Transduction is the transfer of bacterial genes by viruses.
16. In generalized transduction any host DNA fragment can be packaged in a virus capsid and transferred to a recipient (**figure 13.19**).
17. Temperate phages carry out specialized transduction by incorporating bacterial genes during prophage induction and then donating those genes to another bacterium (**figure 13.20**).
18. The bacterial genome can be mapped by following the order of gene transfer during Hfr conjugation (**figure 13.22**); transformational and transductional mapping techniques also may be used.
19. When two viruses simultaneously enter a host cell, their chromosomes can undergo recombination.
20. Virus genomes are mapped by recombination and heteroduplex mapping techniques.

Key Terms

abortive transductants *309*
bacteriocin *297*
competent *305*
composite transposons *298*
conjugation *302*
conjugative plasmid *294*
conjugative transposon *301*
crossing-over *292*
curing *294*
endogenote *294*
episome *294*
exogenote *294*
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transposase *298*
transposition *298*
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virulence plasmids *297*

Questions for Thought and Review

1. How does recombination in procaryotes differ from that in most eucaryotes?
2. Distinguish between plasmids, transposons, and temperate phages.
3. How might one demonstrate the presence of a plasmid in a host cell?
4. What effect would you expect the existence of transposable elements and plasmids to have on the rate of microbial evolution? Give your reasoning.
5. How do multiple drug resistant plasmids often arise?
6. Suppose that you carried out a U-tube experiment with two auxotrophs and discovered that recombination was not blocked by the filter but was stopped by treatment with deoxyribonuclease. What gene transfer process is responsible? Why would it be best to use double or triple auxotrophs in this experiment?
7. List the similarities and differences between conjugation, transformation, and transduction.
8. How might one tell whether a recombination process was mediated by generalized or specialized transduction?
9. Why doesn't a cell lyse after successful transduction with a temperate phage?
10. Describe how you would precisely locate the *recA* gene and show that it was between 58 and 58.5 minutes on the *E. coli* chromosome.

Critical Thinking Questions

1. Diagram a double crossover event and a single crossover event. Which is more infrequent and why? Suggest experiments in which you would use one or the other event and what types of genetic markers you would employ. What kind of recognition features and catalytic capabilities would the recombination machinery need to possess?
2. Suppose that transduction took place when a U-tube experiment was conducted. How would you confirm that something like a virus was passed through the filter and transduced the recipient?
3. What would be the evolutionary advantage of having a period of natural "competence" in a bacterial life cycle? What would be possible disadvantages?

Additional Reading

Chapters 11 and 12 references also should be consulted, particularly the introductory and advanced textbooks.

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PART V

DNA Technology and Genomics

CHAPTER 14

Recombinant DNA Technology

Four streaks of *E. coli* glow different colors because they contain different cloned luciferase genes.

Outline

Concepts

1. Genetic engineering makes use of recombinant DNA technology to fuse genes with vectors and then clone them in host cells. In this way large quantities of isolated genes and their products can be synthesized.
2. The production of recombinant DNA molecules depends on the ability of restriction endonucleases to cleave DNA at specific sites.
3. Plasmids, bacteriophages and other viruses, and cosmids are used as vectors. They can replicate within a host cell while carrying foreign DNA and possess phenotypic traits that allow them to be detected.
4. Genetic engineering is already making substantial contributions to biological research, medicine, industry, and agriculture. Future benefits are probably much greater.
5. Genetic engineering also is accompanied by potential problems in such areas as safety, the ethics of its use with human subjects, environmental impact, and biological warfare.

The recombinant DNA breakthrough has provided us with a new and powerful approach to the questions that have intrigued and plagued man for centuries.

—*Paul Berg*

Chapters 12 and 13 introduce the essentials of microbial genetics. This chapter focuses on the practical applications of microbial genetics and the technology arising from it.

Although human beings have been altering the genetic makeup of organisms for centuries by selective breeding, only recently has the direct manipulation of DNA been possible. The deliberate modification of an organism's genetic information by directly changing its nucleic acid genome is called **genetic engineering** and is accomplished by a collection of methods known as **recombinant DNA technology.** First, the DNA responsible for a particular phenotype is identified and isolated. Once purified, the gene or genes are fused with other pieces of DNA to form recombinant DNA molecules. These are propagated (gene cloning) by insertion into an organism that need not even be in the same kingdom as the original gene donor. Recombinant DNA technology opens up totally new areas of research and applied biology. Thus it is an essential part of **biotechnology,** which is now experiencing a stage of exceptionally rapid growth and development. Although the term has several definitions, in this text biotechnology refers to those processes in which living organisms are manipulated, particularly at the molecular genetic level, to form useful products. The promise for medicine, agriculture, and industry is great; yet the potential risks of this technology are not completely known and may be considerable. Biotechnology and industrial microbiology (chapter 42)

Recombinant DNA technology is very much the result of several key discoveries in microbial genetics. The first section briefly reviews some landmarks in the development of recombinant technology (**table 14.1**).

Table 14.1 Some Milestones in Biotechnology and Recombinant DNA Technology

1958	DNA polymerase purified
1970	A complete gene synthesized in vitro
	Discovery of the first sequence-specific restriction endonuclease and the enzyme reverse transcriptase
1972	First recombinant DNA molecules generated
1973	Use of plasmid vectors for gene cloning
1975	Southern blot technique for detecting specific DNA sequences
1976	First prenatal diagnosis using a gene-specific probe
1977	Methods for rapid DNA sequencing
	Discovery of "split genes" and somatostatin synthesized using recombinant DNA
1978	Human genomic library constructed
1979	Insulin synthesized using recombinant DNA
	First human viral antigen (hepatitis B) cloned
1981	Foot-and-mouth disease viral antigen cloned
	First monoclonal antibody-based diagnostic kit approved for use
1982	Commercial production by *E. coli* of genetically engineered human insulin
	Isolation, cloning, and characterization of a human cancer gene
	Transfer of gene for rat growth hormone into fertilized mouse eggs
1983	Engineered Ti plasmids used to transform plants
1985	Tobacco plants made resistant to the herbicide glyphosate through insertion of a cloned gene from *Salmonella*
	Development of the polymerase chain reaction technique
1987	Insertion of a functional gene into a fertilized mouse egg cures the shiverer mutation disease of mice, a normally fatal genetic disease
1988	The first successful production of a genetically engineered staple crop (soybeans)
	Development of the gene gun
1989	First field test of a genetically engineered virus (a baculovirus that kills cabbage looper caterpillars)
1990	Production of the first fertile corn transformed with a foreign gene (a gene for resistance to the herbicide bialaphos)
1991	Development of transgenic pigs and goats capable of manufacturing proteins such as human hemoglobin
	First test of gene therapy on human cancer patients
1994	The Flavr Savr tomato introduced, the first genetically engineered whole food approved for sale
	Fully human monoclonal antibodies produced in genetically engineered mice
1995	*Haemophilus influenzae* genome sequenced
1996	*Methanococcus jannaschii* and *Saccharomyces cerevisiae* genomes sequenced
1997	Human clinical trials of antisense drugs and DNA vaccines begun
	E. coli genome sequenced
1998	First cloned mammal (the sheep Dolly)

14.1 Historical Perspectives

Recombinant DNA is DNA with a new sequence formed by joining fragments from two or more different sources. One of the first breakthroughs leading to recombinant DNA (rDNA) technology was the discovery in the late 1960s by Werner Arber and Hamilton Smith of microbial enzymes that make cuts in double-stranded DNA. These enzymes recognize and cleave specific sequences about 4 to 8 base pairs long and are known as **restriction enzymes** or restriction endonucleases (**figure 14.1**). They normally protect the host cell by destroying phage DNA after its entrance. Cells protect their own DNA from restriction enzymes by methylating nucleotides in the sites that these enzymes recognize. Incoming foreign DNA is not methylated at the same sites and often is cleaved by host restriction enzymes. There are three general types of restriction enzymes. Types I and III cleave DNA away from recognition sites. Type II restriction endonucleases cleave DNA at specific recognition sites. The type II enzymes can be used to prepare DNA fragments containing specific genes or portions of genes. For example, the restriction enzyme *Eco*RI, isolated by Herbert Boyer in 1969 from *E. coli,* cleaves the DNA between G and A in the base sequence GAATTC (**figure 14.2**). Note that in the double-stranded condition, the base sequence GAATTC will base pair with the same sequence running in the opposite direction. *Eco*RI therefore cleaves both DNA strands between the G and the A. When the two DNA fragments separate, they often contain single-stranded complementary ends, known as sticky ends or cohesive ends. There are hundreds of restriction

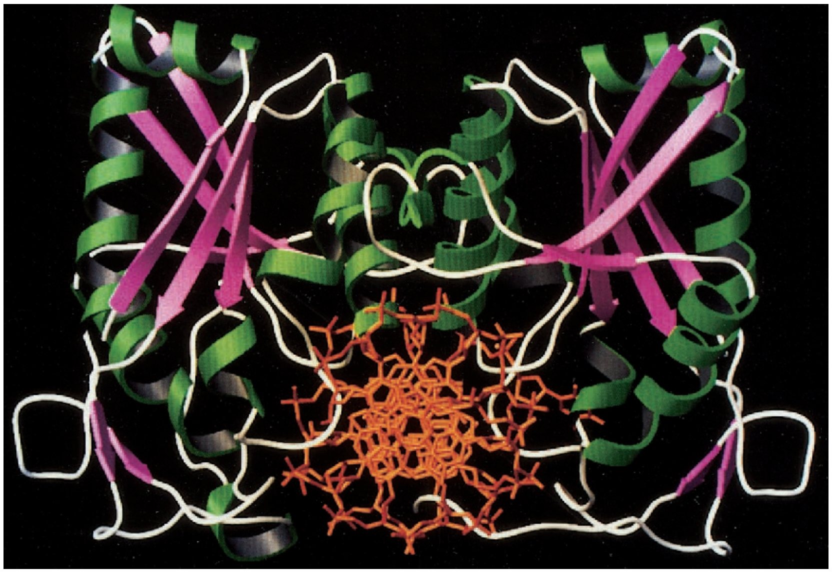


Figure 14.1 Restriction Endonuclease Binding to DNA. The structure of *Bam*HI binding to DNA viewed down the DNA axis. The enzyme's two subunits lie on each side of the DNA double helix. The α-helices are in green, the β conformations in purple to red, and DNA is in orange.

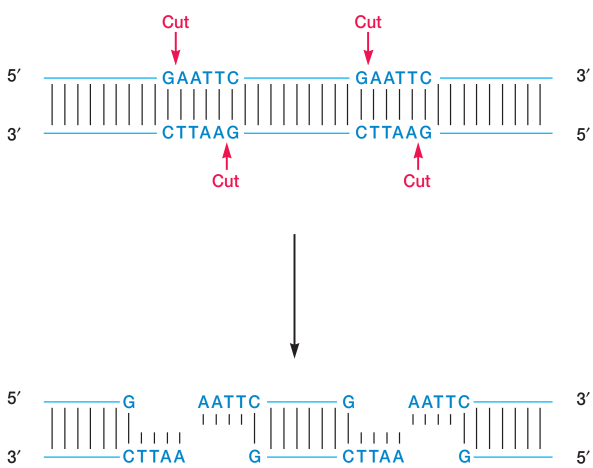


Figure 14.2 Restriction Endonuclease Action. The cleavage catalyzed by the restriction endonuclease *Eco*RI. The enzyme makes staggered cuts on the two DNA strands to form sticky ends.

enzymes that recognize many different specific sequences (**table 14.2**). Each restriction enzyme name begins with three letters, indicating the bacterium producing it. For example, *Eco*RI is obtained from *E. coli,* whereas *Bam*HI comes from *Bacillus amyloliquefaciens H,* and *Sal*I from *Streptomyces albus.* Restriction and phages (p. 386)

In 1970 Howard Temin and David Baltimore independently discovered the enzyme reverse transcriptase that retroviruses use to produce DNA copies of their RNA genome. This enzyme can be used to construct a DNA copy, called **complementary DNA (cDNA),** of any RNA (**figure 14.3**). Thus genes or major portions of genes can be synthesized from mRNA. Reverse transcriptase and retroviruses (p. 407)

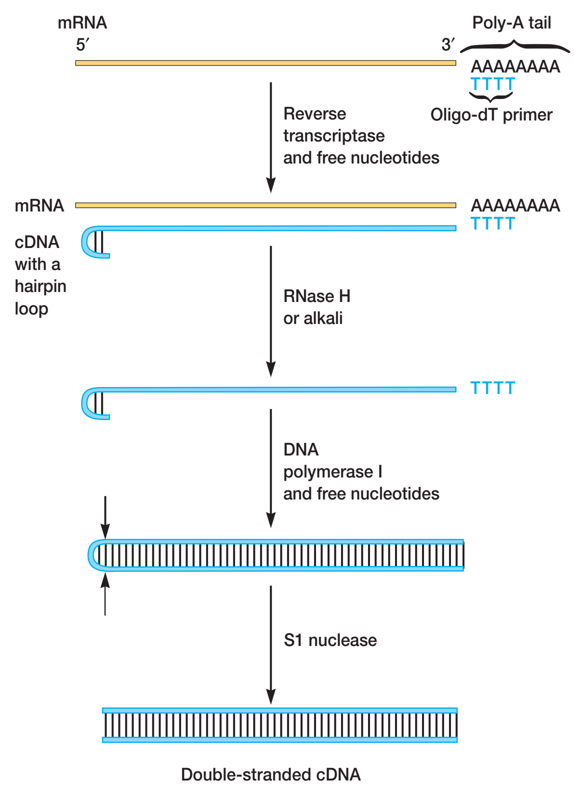


Figure 14.3 The Synthesis of Double-Stranded cDNA from mRNA. This figure briefly summarizes one procedure for synthesis of cDNA. Reverse transcriptase synthesizes a DNA copy of the mRNA using an oligo-dT primer. The RNA of the resulting RNA-DNA hybrid is degraded to produce single-stranded DNA with a hairpin loop at one end. DNA polymerase I then converts this to a double-stranded form, and S1 nuclease nicks the DNA at the points indicated by the two arrows on the left to generate the final DNA copy.

Table 14.2 Some Type II Restriction Endonucleases and Their Recognition Sequences

Enzyme	Microbial Source	Recognition Sequence[a]	End Produced[b]
*Alu*I	*Arthrobacter luteus*	5′—A—G↓C—T—3′ 3′—T—C↑G—A—5′	C—T—3′ G—A—5′
*Bam*HI	*Bacillus amyloliquefaciens H*	5′—G↓G—A—T—C—C—3′ 3′—C—C—T—A—G↑G—5′	G—A—T—C—C—3′ G—5′
*Eco*RI	*Escherichia coli*	5′—G↓A—A—T—T—C—3′ 3′—C—T—T—A—A↑G—5′	A—A—T—T—C—3′ G—5′
*Hae*III	*Haemophilus aegyptius*	5′—G—G↓C—C—3′ 3′—C—C↑G—G—5′	C—C—3′ G—G—5′
*Hind*III	*Haemophilus influenzae* b	5′—A↓A—G—C—T—T—3′ 3′—T—T—C—G—A↑A—5′	A—G—C—T—T—3′ A—5′
*Not*I	*Nocardia otitidis-caviarum*	5′—G—C↓G—G—C—C—G—C—3′ 3′—C—G—C—C—G—G↑C—G—5′	G—G—C—C—G—C—3′ C—G—5′
*Pst*I	*Providencia stuartii*	5′—C—T—G—C—A↓G—3′ 3′—G↑A—C—G—T—C—5′	G—3′ A—C—G—T—C—5′
*Sal*I	*Streptomyces albus*	5′—G↓T—C—G—A—C—3′ 3′—C—A—G—C—T↑G—5′	T—C—G—A—C—3′ G—5′

[a]The arrows indicate the sites of cleavage on each strand.

[b]Only the end of the right-hand fragment is shown.

The next advance came in 1972, when David Jackson, Robert Symons, and Paul Berg reported that they had successfully generated recombinant DNA molecules. They allowed the sticky ends of fragments to anneal—that is, to base pair with one another—and then covalently joined the fragments with the enzyme DNA ligase. Within a year, plasmid **vectors,** or carriers of foreign DNA fragments during gene cloning, had been developed and combined with foreign DNA (**figure 14.4**). The first such recombinant plasmid capable of being replicated within a bacterial host was the pSC101 plasmid constructed by Stanley Cohen and Herbert Boyer in 1973 (SC in the plasmid name stands for *S*tanley *C*ohen).

In 1975 Edwin M. Southern published a procedure for detecting specific DNA fragments so that a particular gene could be isolated from a complex DNA mixture. The **Southern blotting technique** depends on the specificity of base complementarity in nucleic acids (**figure 14.5**). DNA fragments are first separated by size with agarose gel electrophoresis. The fragments are then denatured (rendered single stranded) and transferred to a nitrocellulose filter or nylon membrane so that each fragment is firmly bound to the filter at the same position as on the gel. Originally the transfer occurred when buffer flowed through the gel and the membrane as shown in figure 14.5. The negatively charged DNA fragments also can be electrophoresed from the gel onto the blotting membrane. The filter is bathed with solution containing a radioactive **probe,** a piece of labeled nucleic acid that hybridizes with complementary DNA fragments and is used to locate them. Those fragments complementary to the probe become radioactive and are readily detected by **autoradiography.** In this technique a sheet of photographic film is placed over the filter for several hours and then developed. The film is exposed and becomes dark everywhere a radioactive fragment is located because the energy released by the isotope causes the formation of dark-silver grains. Using Southern blotting, one can detect and isolate fragments with any desired sequence from a complex mixture.

More recently, nonradioactive probes to detect specific DNAs have been developed. In one approach the DNA probe is linked to an enzyme such as horseradish peroxidase. After the enzyme-DNA probe has bound to a DNA fragment on the filter, the substrate luminol that will emit light when acted on by the peroxidase is added. The chemiluminescent probe is detected by exposing the filter to

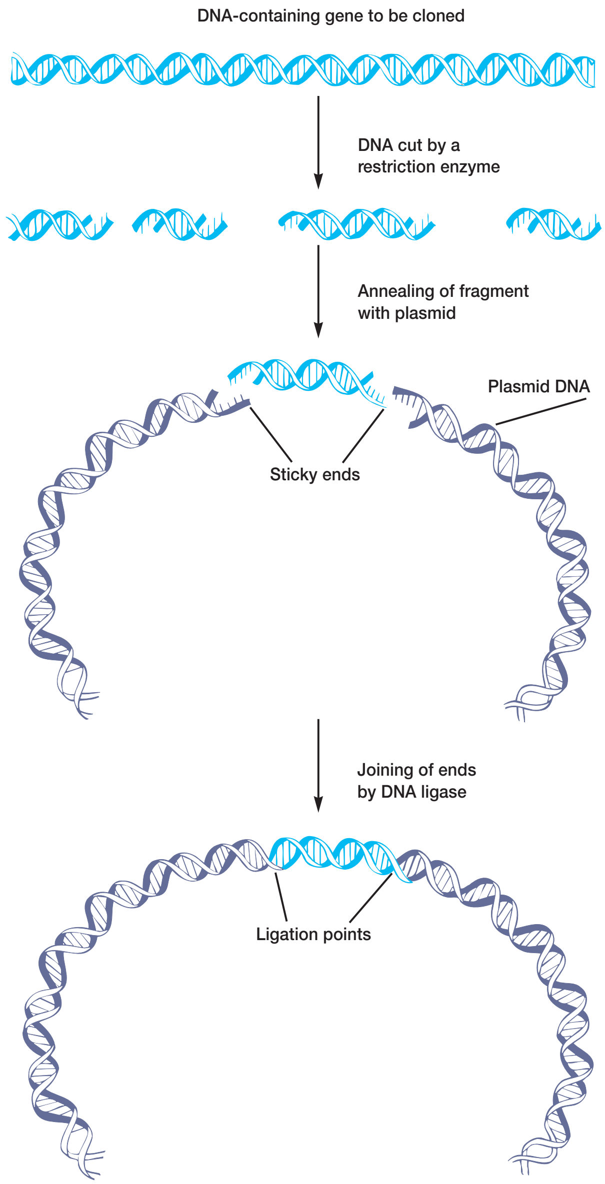


Figure 14.4 Recombinant Plasmid Construction. The general procedure used in constructing recombinant plasmid vectors is shown here.

photographic film for about 20 minutes. A second technique makes use of the vitamin biotin. A biotin-DNA probe is detected by incubating the filter with either the protein avidin or a similar bacterial protein, streptavidin. The protein specifically attaches to biotin, and is visualized with a special reagent containing biotin complexed with the enzyme alkaline phosphatase (streptavidin also can be directly attached to the enzyme). The bands with the probe appear blue. These nonradioactive techniques are more rapid and safer than using radioisotopes. On the other hand, they may be less sensitive than radioactively labeled probes.

By the late 1970s techniques for easily sequencing DNA, synthesizing oligonucleotides, and expressing eucaryotic genes in procaryotes had also been developed. These techniques were then used to solve practical problems (table 14.1). The following sections describe how the previously discussed techniques and others are used in genetic engineering.

1. Define or describe restriction enzyme, sticky end, cDNA, vector, Southern blotting, probe, and autoradiography.

14.2 Synthetic DNA

Oligonucleotides [Greek *oligo,* few or scant] are short pieces of DNA or RNA between about 2 and 20 or 30 nucleotides long. The ability to synthesize DNA oligonucleotides of known sequence is extremely useful. For example, DNA probes can be synthesized and DNA fragments can be prepared for use in molecular techniques such as PCR (p. 326). DNA structure (pp. 231–33)

DNA oligonucleotides are synthesized by a stepwise process in which single nucleotides are added to the end of the growing chain (**figure 14.6**). The 3′ end of the chain is attached to a solid support such as a silica gel particle. A DNA synthesizer or "gene machine" carries out the solid-phase synthesis. A specially activated nucleotide derivative is added to the 5′ end of the chain in a series of steps. At the end of an addition cycle, the growing chain is separated from the reaction mixture by filtration or centrifugation. The process is then repeated to attach another nucleotide. It takes about 40 minutes to add a nucleotide to the chain, and chains as large as 50 to 100 nucleotides can be synthesized.

Advances in DNA synthetic techniques have accelerated progress in the study of protein function. One of the most effective ways of studying the relationship of protein structure to function is by altering a specific part of the protein and observing functional changes. In the past this has been accomplished either by chemically modifying individual amino acids or by inducing mutations in the gene coding for the protein under study. There are problems with these two approaches. Chemical modification of a protein is not always specific; several amino acids may be altered, not just the one desired. It is not always possible to produce the proper mutation in the desired gene location. Recently these difficulties have been overcome with a technique called **site-directed mutagenesis.**

In site-directed mutagenesis an oligonucleotide of about 20 residues that contains the desired sequence change is synthesized. The altered oligonucleotide with its artificially mutated sequence is now allowed to bind to a single-stranded copy of the complete gene (**figure 14.7**). DNA polymerase is added to the gene-primer complex. The polymerase extends the primer and replicates the remainder of the target gene to produce a new gene copy with the desired mutation. If the gene is attached to a single-stranded DNA bacteriophage (*see pp. 372–74 and 388*) such as the M13 phage, it can be introduced into a host bacterium and cloned using the techniques to be described shortly. This will yield large quantities of the mutant protein for study of its function.

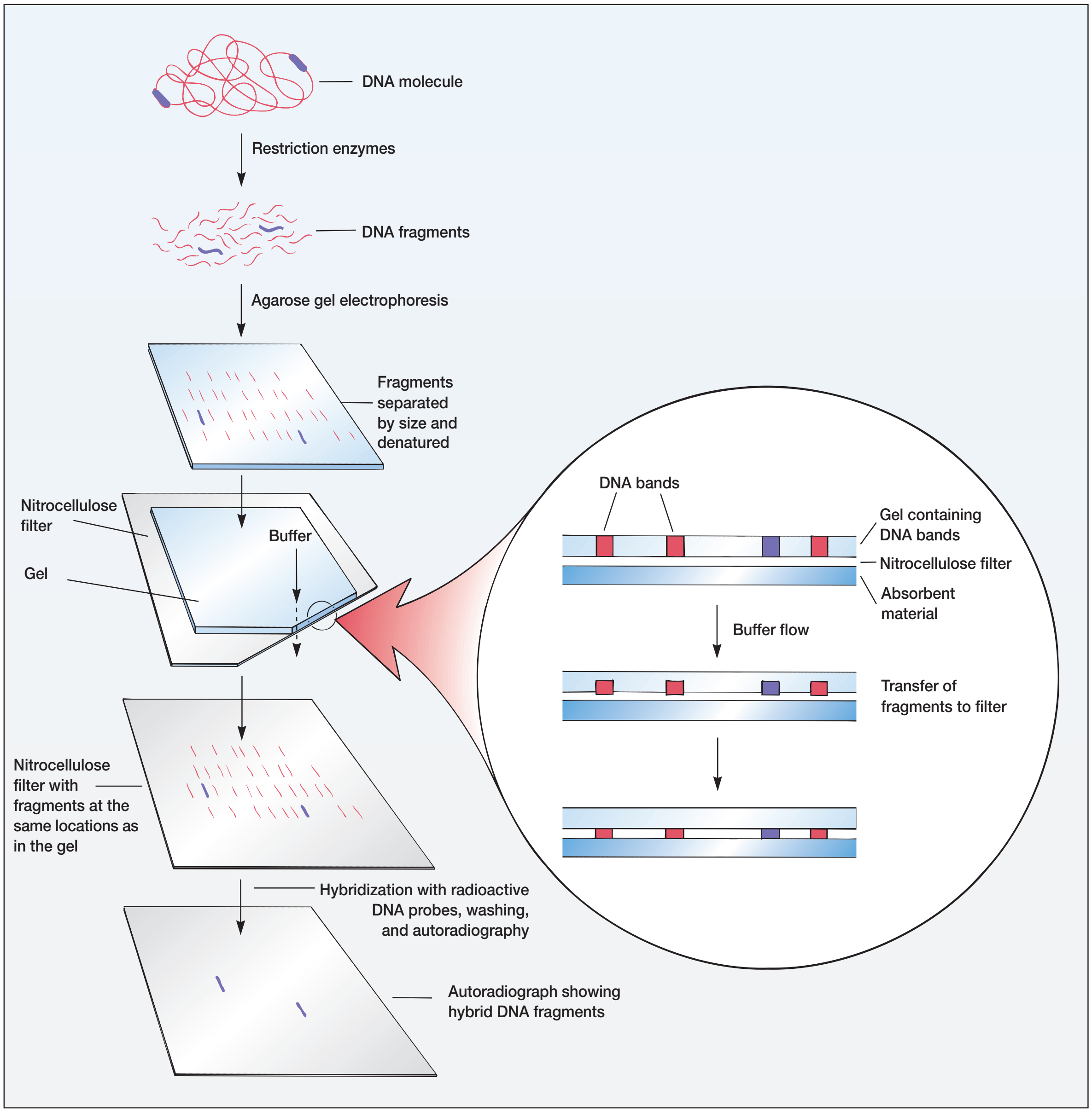


Figure 14.5 The Southern Blotting Technique. The insert illustrates how buffer flow transfers DNA bands to the nitrocellulose filter. The fragments also can be transferred by electrophoresis. See text for further details.

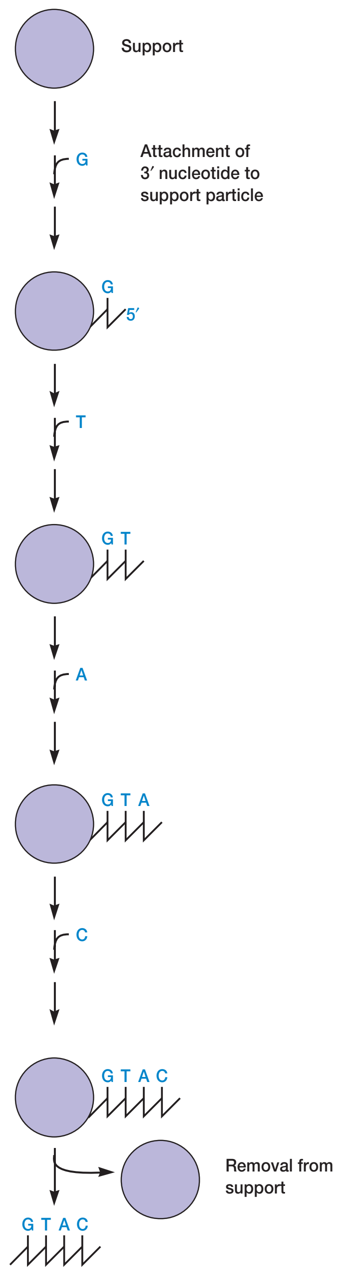


Figure 14.6 The Synthesis of a DNA Oligonucleotide. During each cycle, the DNA synthesizer adds an activated nucleotide (A, T, G, or C) to the growing end of the chain. At the end of the process, the oligonucleotide is removed from its support.

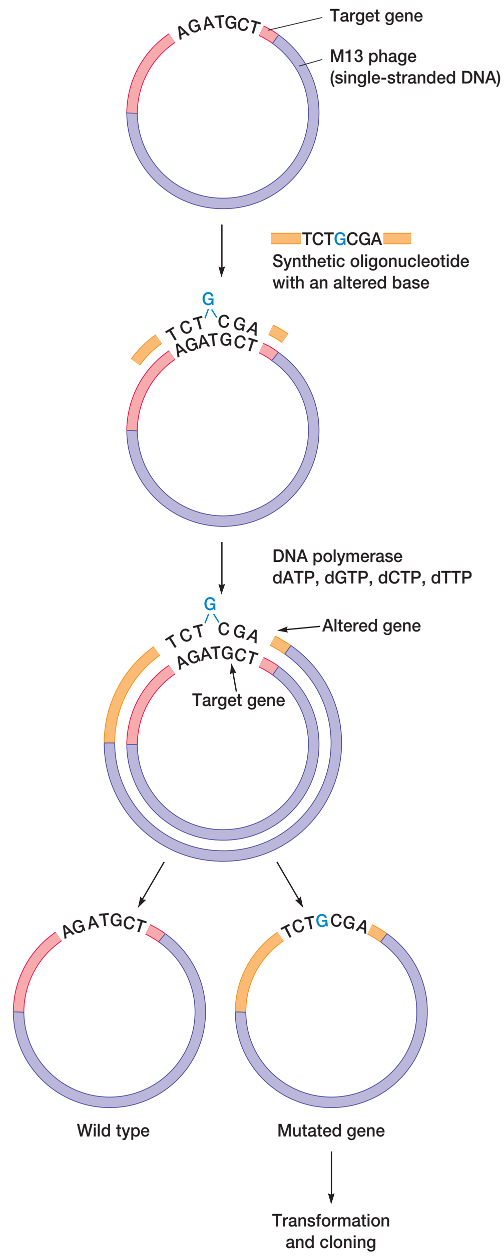


Figure 14.7 Site-Directed Mutagenesis. A synthetic oligonucleotide is used to add a specific mutation to a gene. See text for details.

14.3 The Polymerase Chain Reaction

Between 1983 and 1985 Kary Mullis developed a new technique that made it possible to synthesize large quantities of a DNA fragment without cloning it. Although this chapter emphasizes recombinant DNA technology, the **polymerase chain reaction** or **PCR technique** will be introduced here because of its great practical importance and impact on biotechnology.

Figure 14.8 outlines how the PCR technique works. Suppose that one wishes to make large quantities of a particular DNA sequence. The first step is to synthesize fragments with sequences identical to those flanking the targeted sequence. This is readily accomplished with a DNA synthesizer machine as previously described. These synthetic oligonucleotides are usually about 20 nucleotides long and serve as primers for DNA synthesis. The reaction mix contains the target DNA, a very large excess of the desired primers, a thermostable DNA polymerase, and four deoxyribonucleoside triphosphates. The PCR cycle itself takes place in three steps. First, the target DNA containing the sequence to be amplified is heat denatured to separate its complementary strands (step 1). Normally the target DNA is between 100 and 5,000 base pairs in length. Next, the temperature is lowered so that the primers can hydrogen bond or anneal to the DNA on both sides of the target sequence (step 2). Because the primers are present in excess, the targeted DNA strands normally anneal to the primers rather than to each other. Finally, DNA polymerase extends the primers and synthesizes copies of the target DNA sequence using the deoxyribonucleoside triphosphates (step 3). Only polymerases able to function at the high temperatures employed in the PCR technique can be used. Two popular enzymes are the Taq polymerase from the thermophilic bacterium *Thermus aquaticus* and the Vent polymerase from *Thermococcus litoralis*. At the end of one cycle, the targeted sequences on both strands have been copied. When the three-step cycle is repeated (figure 14.8), the four strands from the first cycle are copied to produce eight fragments. The third cycle yields 16 products. Theoretically, 20 cycles will produce about one million copies of the target DNA sequence; 30 cycles yield around one billion copies. Pieces ranging in size from less than 100 base pairs to several thousand base pairs in length can be amplified, and only 10 to 100 picomoles of primer are required. The concentration of target DNA can be as low as 10^{-20} to 10^{-15} M (or 1 to 10^5 DNA copies per 100 μl). The whole reaction mixture is often 100 μl or less in volume. DNA sequencing (p. 345); DNA replication (pp. 235–39)

The polymerase chain reaction technique has now been automated and is carried out by a specially designed machine (**figure 14.9**). Currently a PCR machine can carry out 25 cycles and amplify DNA 10^5 times in as little as 57 minutes. During a typical cycle the DNA is denatured at 94°C for 15 seconds, then the primers are annealed and extended (steps 2 and 3) at 68°C for 60 seconds.

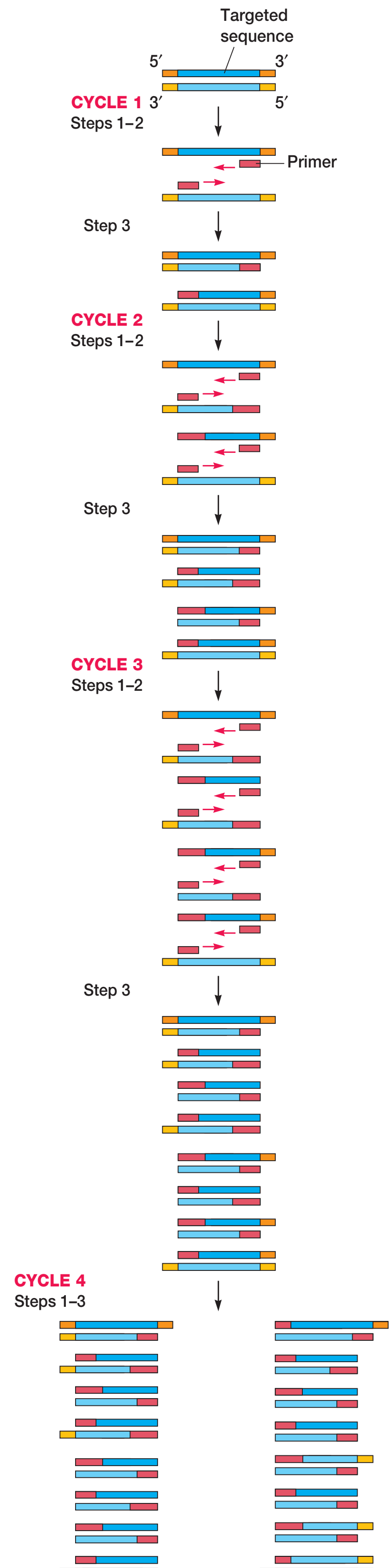


Figure 14.8 The Polymerase Chain Reaction (PCR). In four cycles the targeted sequence has been amplified many times. See text for details.

Figure 14.9 A Modern PCR Machine. PCR machines are now fully automated and microprocessor controlled. They can process up to 96 samples at a time.

PCR technology is improving continually. For example, RNA now can be efficiently used in PCR procedures. The *Tth* DNA polymerase, a recombinant *Thermus thermophilus* DNA polymerase, will transcribe RNA to DNA and then amplify the DNA. Cellular RNAs and RNA viruses may be studied even when the RNA is present in very small amounts (as few as 100 copies can be transcribed and amplified). PCR also can quantitate DNA products without the use of isotopes. This allows one to find the initial amount of target DNA in less than an hour using automated equipment. Quantitative PCR is quite valuable in virology and gene expression studies. As mentioned earlier, the target DNA to be amplified is normally less than about 5,000 base pairs in length. A "long PCR" technique has been developed that will amplify sequences up to 42 kilobases long. It depends on the use of error-correcting polymerases because Taq polymerase is error-prone.

The PCR technique has already proven exceptionally valuable in many areas of molecular biology, medicine, and biotechnology. It can be used to amplify very small quantities of a specific DNA and provide sufficient material for accurately sequencing the fragment or cloning it by standard techniques. PCR-based diagnostic tests for AIDS, Lyme disease, chlamydia, tuberculosis, hepatitis, the human papilloma virus, and other infectious agents and diseases are being developed. The tests are rapid, sensitive, and specific. PCR is particularly valuable in the detection of genetic diseases such as sickle cell anemia, phenylketonuria, and muscular dystrophy. The technique is already having an impact on forensic science where it is being used in criminal cases as a part of DNA fingerprinting technology. It is possible to exclude or incriminate suspects using extremely small samples of biological material discovered at the crime scene.

14.4 Preparation of Recombinant DNA

There are three ways to obtain adequate quantities of a DNA fragment. One can extract all the DNA from an organism, cleave the DNA into fragments, isolate the fragment of interest, and finally clone it. Alternatively, all of the fragments can be cloned by means of a suitable vector, and each clone (the population of identical molecules with a single ancestral molecule) can be tested for the desired gene. One also can directly synthesize the desired DNA fragment as described earlier, and then clone it.

Isolating and Cloning Fragments

Long, linear DNA molecules are fragile and easily sheared into fragments by passing the DNA suspension through a syringe needle several times. DNA also can be cut with restriction enzymes. The resulting fragments are either separated by electrophoresis or first inserted into vectors and cloned.

Agarose or polyacrylamide gels usually are used to separate DNA fragments electrophoretically. In **electrophoresis,** charged molecules are placed in an electrical field and allowed to migrate toward the positive and negative poles. The molecules separate because they move at different rates due to their differences in charge and size. In practice, the fragment mixture is usually placed in wells molded within a sheet of gel (**figure 14.10**). The gel concentration varies with the size of DNA fragments to be separated. Usually 1 to 3% agarose gels or 3 to 20% polyacrylamide gels are used. When an electrical field is generated in the gel, the fragments move through the pores of the gel toward an electrode (negatively charged DNA fragments migrate toward the positive electrode or anode). Each fragment's migration rate is inversely proportional to the log of its molecular weight, and the fragments are separated into size classes (figure 14.10*b*). A simple DNA molecule might yield only a few bands. If the original DNA was very large and complex, staining of the gel would reveal a smear representing an almost continuous gradient in fragment size. The band or section containing the desired fragment is located with the Southern blotting technique (figure 14.5) and removed. Since a band may represent a mixture of several fragments, it is electrophoresed on a gel with a different concentration to separate similarly sized fragments. The location of the pure fragment is determined by Southern blotting, and it is extracted from the gel.

Once fragments have been isolated, they are ligated with an appropriate vector, such as a plasmid (*see section 13.2*), to form a recombinant molecule that can reproduce in a host cell. One of the easiest and most popular approaches is to cut the plasmid and donor DNA with the same restriction enzyme so that identical sticky ends are formed (**figure 14.11**).

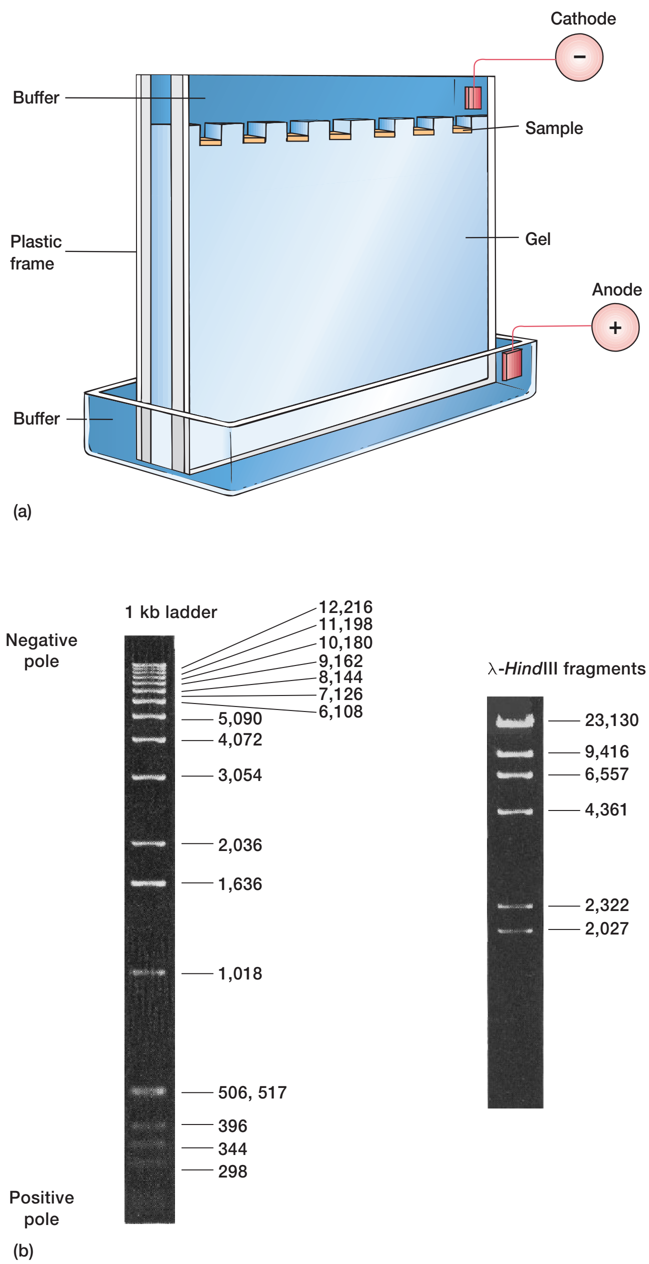


Figure 14.10 Gel Electrophoresis of DNA. (**a**) A diagram of a vertical gel apparatus showing its parts. (**b**) The 1 kilobase ladder is an electrophoretic gel containing a series of DNA fragments of known size. The numbers indicate number of base pairs each fragment contains. The smallest fragments have moved the farthest. The gel on the right shows many of the fragments that arise when lambda phage DNA is digested with the *Hind*III restriction enzyme.

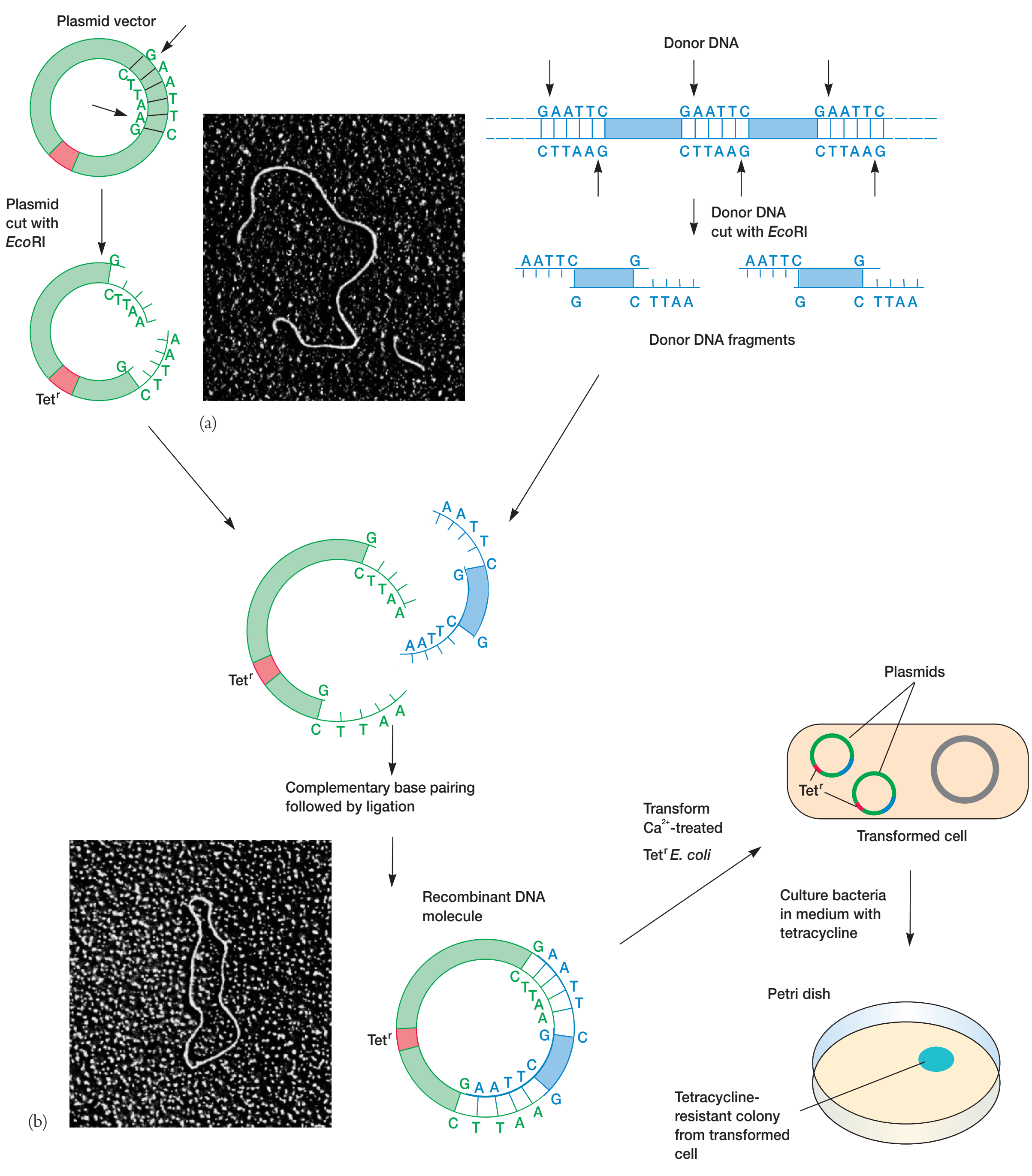


Figure 14.11 Recombinant Plasmid Construction and Cloning. The construction and cloning of a recombinant plasmid vector using an antibiotic resistance gene to select for the presence of the plasmid. The scale of the sticky ends of the fragments and plasmid has been enlarged to illustrate complementary base pairing. (**a**) The electron micrograph shows a plasmid that has been cut by a restriction enzyme and a donor DNA fragment. (**b**) The micrograph shows a recombinant plasmid. See text for details.

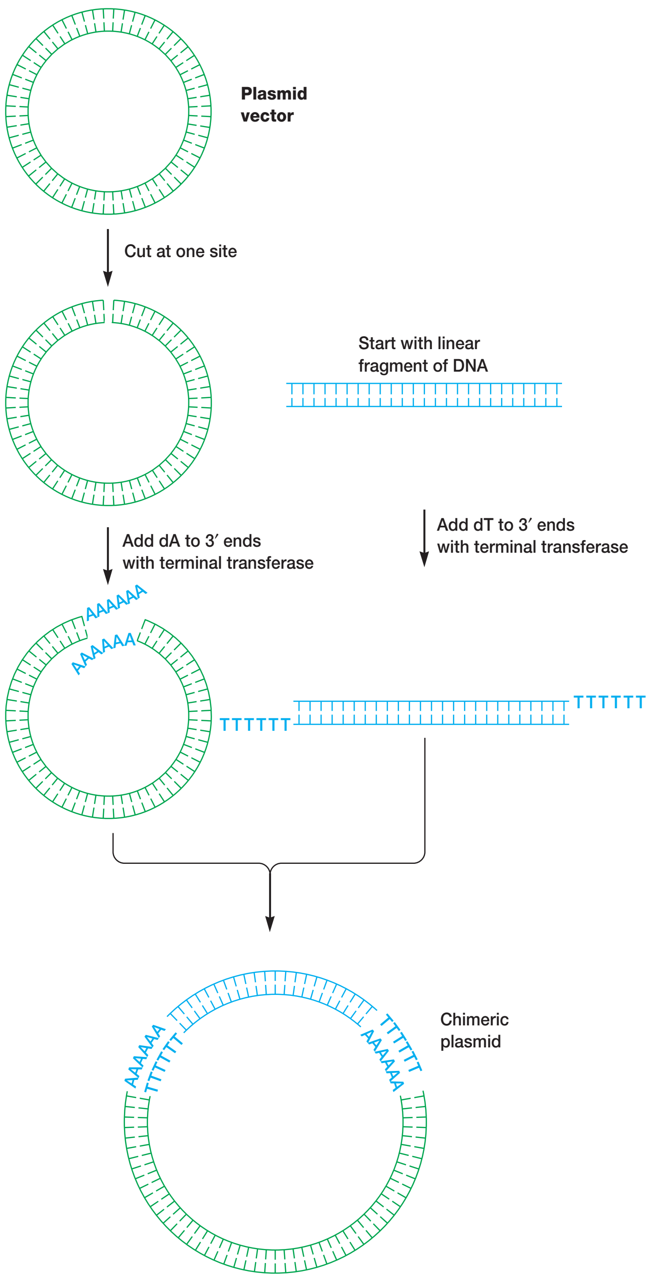


Figure 14.12 Terminal Transferase and the Construction of Recombinant Plasmids. The poly(dA-dT) tailing technique can be used to construct sticky ends on DNA and generate recombinant molecules.

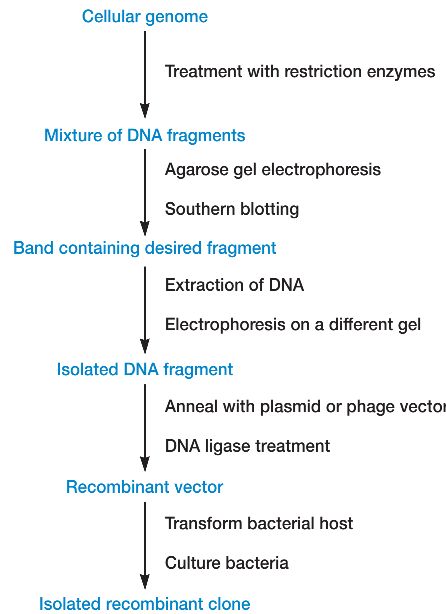


Figure 14.13 Cloning Cellular DNA Fragments. The preparation of a recombinant clone from previously isolated DNA fragments.

After a fragment has annealed with the plasmid through complementary base pairing, the breaks are joined by DNA ligase. A second method for creating recombinant molecules can be used with fragments and vectors lacking sticky ends. After cutting the plasmid and donor DNA, one can add poly(dA) to the 3′ ends of the plasmid DNA, using the enzyme terminal transferase (**figure 14.12**). Similarly, poly(dT) is added to the 3′ ends of the fragments. The ends will now base pair with each other and are joined by DNA ligase to form a recombinant plasmid. Some enzymes (e.g., the *Alu*I restriction enzyme from *Arthrobacter luteus*) cut DNA at the same position on both strands to form a blunt end. Fragments and vectors with blunt ends may be joined by T4 DNA ligase (blunt-end ligation).

The rDNA molecules are cloned by inserting them into bacteria, using transformation or phage injection (*see section 13.5*). Each strain reproduces to yield a population containing a single type of recombinant molecule. The overall process is outlined in **figure 14.13.** The same cloning techniques can be used with DNA fragments prepared using a DNA synthesizer machine.

Although selected fragments can be isolated and cloned as just described, it often is preferable to fragment the whole genome and clone all the fragments by using a vector. Then the desired clone can be identified. To be sure that the complete genome is represented in this collection of clones, called a genomic **library,** more than a thousand transformed bacterial strains must be maintained (the larger the genome, the more clones are needed). Libraries of cloned genes also can be generated using phage lambda as a vector and stored as phage lysates.

It is necessary to identify which clone in the library contains the desired gene. If the gene is expressed in the bacterium, it may be possible to assay each clone for a specific protein. However, a nucleic acid probe is normally employed in identification. The bacteria are replica plated on nitrocellulose paper

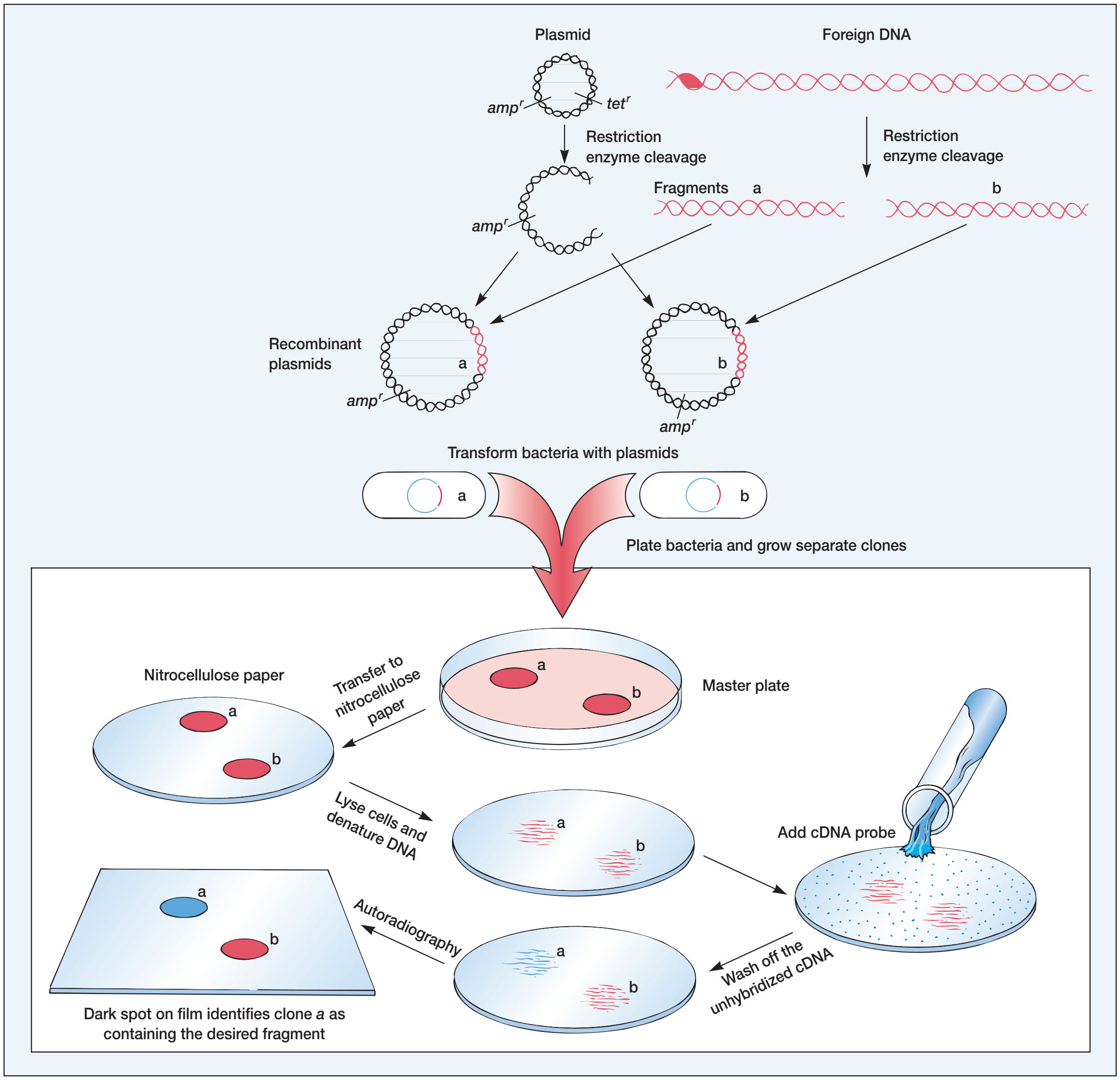


Figure 14.14 Cloning with Plasmid Vectors. The use of plasmid vectors to clone a mixture of DNA fragments. The desired fragment is identified with a cDNA probe. See text for details.

and lysed in place with sodium hydroxide (**figure 14.14**). This yields a pattern of membrane-bound, denatured DNA corresponding to the colony pattern on the agar plate. The membrane is treated with a radioactive probe as in the original Southern blotting method. The radioactive spots identify colonies on the master plate that contain the desired DNA fragment. This approach also is used to analyze a library of cloned lambda phage (**figure 14.15**).

Gene Probes

Success in isolating the desired recombinant clones depends on the availability of a suitable probe. Gene-specific probes are obtained in several ways. Frequently they are constructed with cDNA clones. If the gene of interest is expressed in a specific tissue or cell type, its mRNA is often relatively abundant. For example, reticulocyte mRNA may be enriched in globin mRNA, and pancreatic cells

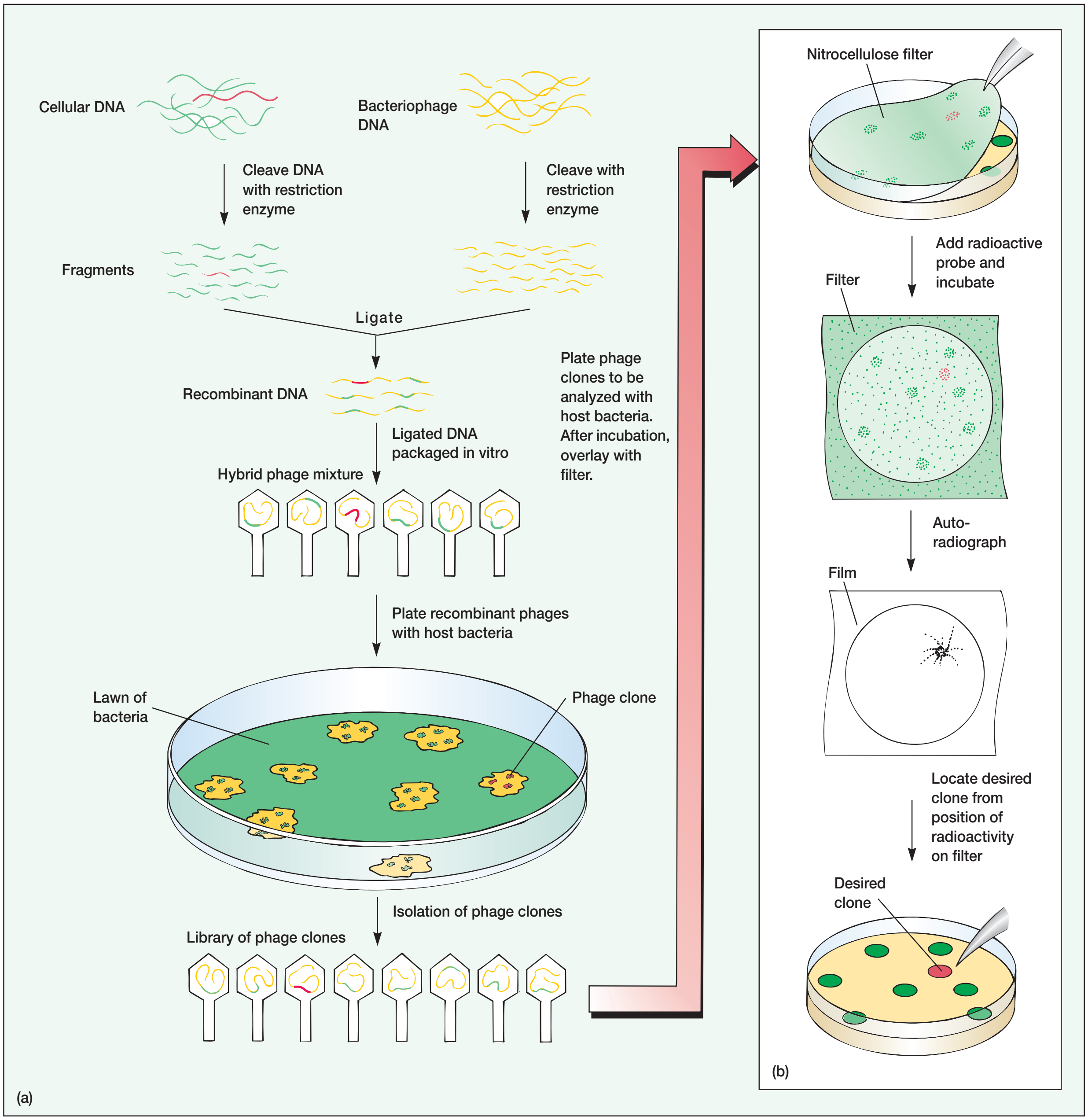


Figure 14.15 The Use of Lambda Phage as a Vector. (**a**) The preparation of a genomic library. Each colony on the bacterial lawn is a recombinant clone carrying a different DNA fragment. (**b**) Detection and cloning of the desired recombinant phage.

Table 14.3 Some Recombinant DNA Cloning Vectors

Type	Vector	Restriction Sequences Present	Features
Plasmid (*E. coli*)	pBR322	*Bam*HI, *Eco*RI, *Hae*III, *Hind*III, *Pst*I, *Sal*I, *Xor*II	Carries genes for tetracycline and ampicillin resistance
Plasmid (yeast–*E. coli* hybrid)	pYe(CEN3)41	*Bam*HI, *Bgl*II, *Eco*RI, *Hind*III, *Pst*I, *Sal*I	Multiplies in *E. coli* or yeast cells
Cosmid (artificially constructed *E. coli* plasmid carrying lambda cos site)	pJC720	*Hind*III	Can be packaged in lambda phage particles for efficient introduction into bacteria; replicates as a plasmid; useful for cloning large DNA inserts
YAC (yeast artificial chromosome)	pYAC	*Sma*I, *Bam*HI	Carries gene for ampicillin resistance; multiplies in *Saccharomyces cerevisiae.*
BAC (bacterial artificial chromosome)	pBAC108L	*Hind*III, *Bam*HI, *Not*I, *Sma*I, and others	Modified F plasmid that can carry 100–300 kb fragments; has a *cosN* site and a chloramphenicol resistance marker
Virus	Charon phage	*Eco*RI, *Hind*III, *Bam*HI, *Sst*I	Constructed using restriction enzymes and a ligase, having foreign DNA as central portion, with lambda DNA at each end; carries β-galactosidase gene; packaged into lambda phage particles; useful for cloning large DNA inserts
Virus	Lambda 1059	*Bam*HI	Will carry large DNA fragments (8–21 kb); recombinant can grow on *E. coli* lysogenic for P2 phage, whereas vector cannot
Virus	M13	*Eco*RI	Single-stranded DNA virus; useful in studies employing single-stranded DNA insert and in producing DNA fragments for sequencing
Plasmid	Ti	*Sma*I, *Hpa*I	Maize plasmid

Adapted from G. D. Elseth and K. D. Baumgartner, *Genetics,* 1984 Benjamin/Cummings Publishing, Menlo Park, CA. Reprinted by permission of the author.

in insulin mRNA. Although mRNA is not available in sufficient quantity to serve as a probe, the desired mRNA species can be converted into cDNA by reverse transcription (figure 14.3). The cDNA copies are purified, spliced into appropriate vectors, and cloned to provide adequate amounts of the required probe.

Probes also can be generated if the gene codes for a protein of known amino acid sequence. Oligonucleotides, about 20 nucleotides or longer, that code for a characteristic amino acid sequence are synthesized. These often are satisfactory probes since they will specifically bind to the gene segment coding for the desired protein.

Sometimes previously cloned genes or portions of genes may be used as probes. This approach is effective when there is a reasonable amount of similarity between the nucleotide sequences of the two genes. Probes also can be generated by the polymerase chain reaction.

After construction, the probe is labeled to aid detection. Often ^{32}P is added to both DNA strands so that the radioactive strands can be located with autoradiography. Nonradioactively labeled probes may also be used.

Isolating and Purifying Cloned DNA

After the desired clone of recombinant bacteria or phages has been located with a probe, it can be picked from the master plate and propagated. The recombinant plasmid or phage DNA is then extracted and further purified when necessary. The DNA fragment is cut out of the plasmid or phage genome by means of restriction enzymes and separated from the remaining DNA by electrophoresis.

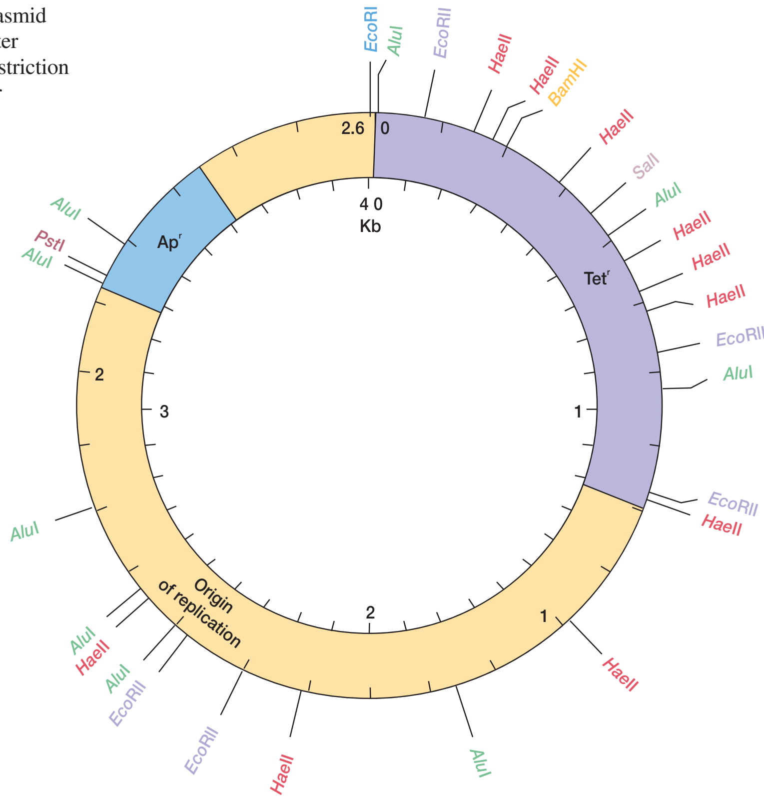
Clearly, DNA fragments can be isolated, purified, and cloned in several ways. Regardless of the exact approach, a key to successful cloning is choosing the right vector. The next section considers types of cloning vectors and their uses.

1. How are oligonucleotides synthesized? What is site-directed mutagenesis?
2. Briefly describe the polymerase chain reaction technique. What is its importance?
3. What is electrophoresis and how does it work?
4. Describe three ways in which a fragment can be covalently attached to vector DNA.
5. Outline in detail two different ways to isolate and clone a specific gene. What is a genomic library?
6. How are gene-specific probes obtained?

14.5 Cloning Vectors

There are four major types of vectors: plasmids, bacteriophages and other viruses, cosmids, and artificial chromosomes (**table 14.3**). Each type has its own advantages. Plasmids are the easiest to work with; rDNA phages and other viruses are more conveniently stored for long periods; larger pieces of DNA can be cloned with cosmids and artificial chromosomes. Besides these major types, there also are vectors designed for a specific function. For example, shuttle

Figure 14.16 The pBR322 Plasmid. A map of the *E. coli* plasmid pBR322. The map is marked off in units of 1×10^5 daltons (outer circle) and 0.1 kilobases (inner circle). The locations of some restriction enzyme sites are indicated. The plasmid has resistance genes for ampicillin (Ap^r) and tetracycline (Tet^r).



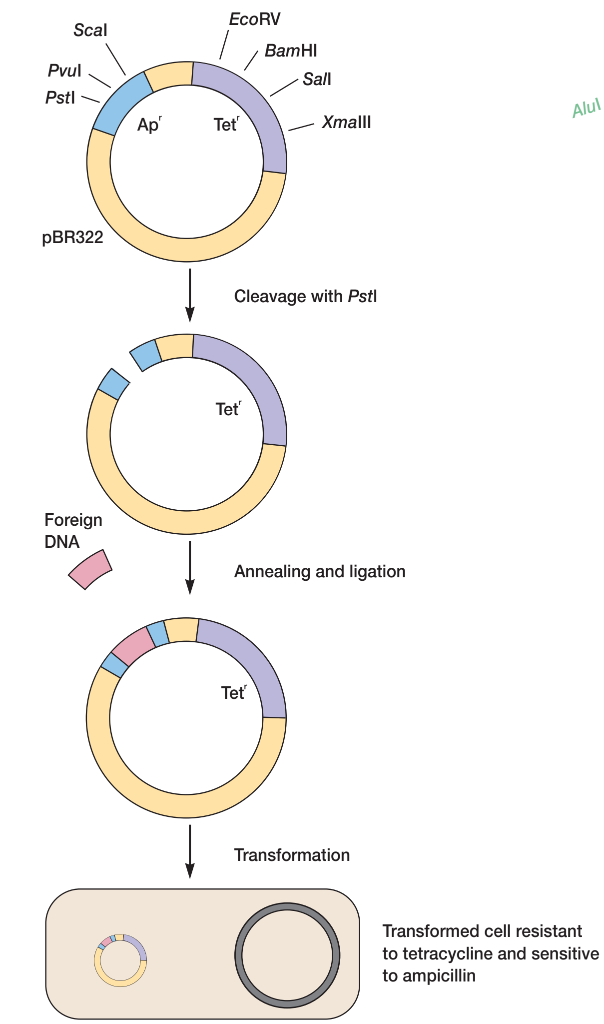


Figure 14.17 Detection of Recombinant Plasmids. The use of antibiotic resistance genes to detect the presence of recombinant plasmids. Because foreign DNA has been inserted into the ampicillin resistance gene, the recombinant host is only resistant to tetracycline. The restriction enzymes indicated at the top of the figure cleave only one site on the plasmid.

vectors are used in transferring genes between very different organisms and usually contain one replication origin for each host. The pYe type plasmids reproduce in both yeast and *E. coli* and can be used to clone yeast genes in *E. coli.*

All vectors share several common characteristics. They are typically small, well-characterized molecules of DNA. They contain at least one replication origin and can be replicated within the appropriate host, even when they contain "foreign" DNA. Finally, they code for a phenotypic trait that can be used to detect their presence; frequently it is also possible to distinguish parental from recombinant vectors.

Plasmids

Plasmids were the first cloning vectors. They are easy to isolate and purify, and they can be reintroduced into a bacterium by transformation. Plasmids often bear antibiotic resistance genes, which are used to select their bacterial hosts. A recombinant plasmid containing foreign DNA often is called a **chimera,** after the Greek mythological monster that had the head of a lion, the tail of a dragon, and the body of a goat. One of the most widely used plasmids is pBR322. The biology of plasmids (pp. 294–97)

Plasmid pBR322 has both resistance genes for ampicillin and tetracycline and many restriction sites (**figure 14.16**). Several of these restriction sites occur only once on the plasmid and are located within an antibiotic resistance gene. This arrangement aids detection of recombinant plasmids after transformation (**figure 14.17**). For example, if foreign DNA is inserted into the ampicillin resistance gene, the plasmid will no longer confer resistance to ampicillin. Thus tetracycline-resistant transformants that lack ampicillin resistance contain a chimeric plasmid.

Phage Vectors

Both single- and double-stranded phage vectors have been employed in recombinant DNA technology. For example, lambda phage derivatives are very useful for cloning and can carry fragments up to about 45 kilobases in length. The genes for lysogeny and integration often are nonfunctional and may be deleted to make room for the foreign DNA. The modified phage genome also contains restriction sequences in areas that will not disrupt replication. After insertion of the foreign DNA into the modified lambda vector chromosome, the recombinant phage genome is packaged into viral capsids and can be used to infect host *E. coli* cells (**figure 14.15**). These vectors are often used to generate genomic libraries. *E. coli* also can be directly transformed with recombinant lambda DNA and produce phages. However, this approach is less efficient than the use of complete phage particles. The process is sometimes called **transfection.** Phages other than lambda also are used as vectors. For example, fragments as large as 95 kilobases can be carried by the P1 bacteriophage. The biology of bacteriophages (chapter 17)

Cosmids

Cosmids are plasmids that contain lambda phage cos sites and can be packaged into phage capsids. The lambda genome contains a recognition sequence called a cos site (or cohesive end) at each end. When the genome is to be packaged in a capsid, it is cleaved at one cos site and the linear DNA is inserted into the capsid until the second cos site has entered. Thus any DNA inserted between the cos sites is packaged. Cosmids typically contain several restriction sites and antibiotic resistance genes. They are packaged in lambda capsids for efficient injection into bacteria, but they also can exist as plasmids within a bacterial host. As much as 50 kilobases of DNA can be carried in this way.

Artificial Chromosomes

Artificial chromosomes are popular because they carry large amounts of genetic material. The **yeast artificial chromosome (YAC)** is one of the most widely used. YACs are stretches of DNA that contain all the elements required to propagate a chromosome in yeast: a replication origin, the centromere required to segregate chromatids into daughter cells, and two telomeres to mark the ends of the chromosome. They will also have restriction enzyme sites and genetic markers so that they can be traced and selected. Cleavage of a YAC with the proper restriction enzyme such as *Sma*I will open it up and allow the insertion of a piece of foreign DNA between the centromere and a telomere. In this way YACs containing DNA fragments between 100 and 2,000 kilobases in size can be placed in *Saccharomyces cerevisiae* cells and will be replicated along with the true chromosomes. The **bacterial artificial chromosome (BAC)** is an increasingly popular alternative cloning vector. BACs are cloning vectors based on the *E. coli* F-factor plasmid. They contain appropriate restriction enzyme sites and a marker such as chloramphenicol resistance. The modified plasmid is cleaved at a restriction site, and a foreign DNA fragment up to 300 kilobases in length is attached using DNA ligase. The BAC is reproduced in *E. coli* after insertion by electroporation. This vector is easy to reproduce and manipulate and does not undergo recombination as readily as YACs (which means that the insert is less likely to be rearranged). Because they can carry such large fragments of DNA, artificial chromosomes have been particularly useful in genome sequencing.

1. Define shuttle vector, chimera, cosmid, transfection, yeast, artificial chromosome, and bacterial artificial chromosome.
2. Describe how each of the major types of vectors is used in genetic engineering. Give an advantage of each.
3. How can the presence of two antibiotic resistance genes be used to detect recombinant plasmids?

14.6 Inserting Genes into Eucaryotic Cells

Because of its practical importance, much effort has been devoted to the development of techniques for inserting genes into eucaryotic cells. Some of these techniques have also been used successfully to transform bacterial cells. The most direct approach is the use of microinjection. Genetic material directly injected into animal cells such as fertilized eggs is sometimes stably incorporated into the host genome to produce a **transgenic animal,** one that has gained new genetic information from the acquisition of foreign DNA.

Another effective technique that works with mammalian cells and plant cell protoplasts is **electroporation.** If cells are mixed with a DNA preparation and then briefly exposed to pulses of high voltage (from about 250 to 4,000 V/cm for mammalian cells), the cells take up DNA through temporary holes in the plasma membrane. Some of these cells will be transformed.

One of the most effective techniques is to shoot microprojectiles coated with DNA into plant and animal cells. The **gene gun,** first developed at Cornell University, operates somewhat like a shotgun. A blast of compressed gas shoots a spray of DNA-coated metallic microprojectiles into the cells. The device has been used to transform corn and produce fertile corn plants bearing foreign genes. Other guns use either electrical discharges or high-pressure gas to propel the DNA-coated projectiles. These guns are sometimes called biolistic devices, a name derived from biological and ballistic. They have been used to transform microorganisms (yeast, the mold *Aspergillus,* and the alga *Chlamydomonas*), mammalian cells, and a variety of plant cells (corn, cotton, tobacco, onion, and poplar).

Other techniques also are available. Plants can be transformed with *Agrobacterium* vectors as will be described shortly (p. 340). Viruses increasingly are used to insert desired genes into eucaryotic cells. For example, genes may be placed in a retrovirus (*see p. 407*), which then infects the target cell and integrates a

DNA copy of its RNA genome into the host chromosome. Adenoviruses also can transfer genes to animal cells. Recombinant baculoviruses will infect insect cells and promote the production of many proteins.

14.7 Expression of Foreign Genes in Bacteria

After a suitable cloning vector has been constructed, rDNA enters the host cell, and a population of recombinant microorganisms develops. Most often the host is an *E. coli* strain that lacks restriction enzymes and is *recA*$^{-}$ to reduce the chances that the rDNA will undergo recombination with the host chromosome. *Bacillus subtilis* and the yeast *Saccharomyces cerevisiae* also may serve as hosts. Plasmid vectors enter *E. coli* cells by calcium chloride-induced transformation. Electroporation at 3 to 24 kV/cm is effective with both gram-positive and gram-negative bacteria.

A cloned gene is not always expressed in the host cell without further modification of the recombinant vector. To be transcribed, the recombinant gene must have a promoter that is recognized by the host RNA polymerase. Translation of its mRNA depends on the presence of leader sequences and mRNA modifications that allow proper ribosome binding. These are quite different in eucaryotes and procaryotes, and a procaryotic leader must be provided to synthesize eucaryotic proteins in a bacterium. Finally, introns in eucaryotic genes must be removed because the procaryotic host will not excise them after transcription of mRNA; a eucaryotic protein is not functional without intron removal prior to translation.

The problems of expressing recombinant genes in host cells are largely overcome with the help of special cloning vectors called **expression vectors.** These vectors are often derivatives of plasmid pBR322 and contain the necessary transcription and translation start signals. They also have useful restriction sites next to these sequences so that foreign DNA can be inserted with relative ease. Some expression vectors contain portions of the lac operon and can effectively regulate the expression of the cloned genes in the same manner as the operon.

Somatostatin, the 14-residue hypothalamic polypeptide hormone that helps regulate human growth, provides an example of useful cloning and protein production. The gene for somatostatin was initially synthesized by chemical methods. Besides the 42 bases coding for somatostatin, the polynucleotide contained a codon for methionine at the 5′ end (the N-terminal end of the peptide) and two stop codons at the opposite end. To aid insertion into the plasmid vector, the 5′ ends of the synthetic gene were extended to form single-stranded sticky ends complementary to those formed by the *Eco*RI and *Bam*HI restriction enzymes. A modified pBR322 plasmid was cut with both *Eco*RI and *Bam*HI to remove a part of the plasmid DNA. The synthetic gene was then spliced into the vector by taking advantage of its cohesive ends (**figure 14.18**). Finally, a fragment containing the initial part of the lac operon (including the promoter, operator, ribosome binding site, and much of the β-galactosidase gene) was inserted next to the somatostatin gene. The plasmid now contained the somatostatin gene fused in the proper orientation to the remaining portion of the β-galactosidase gene.

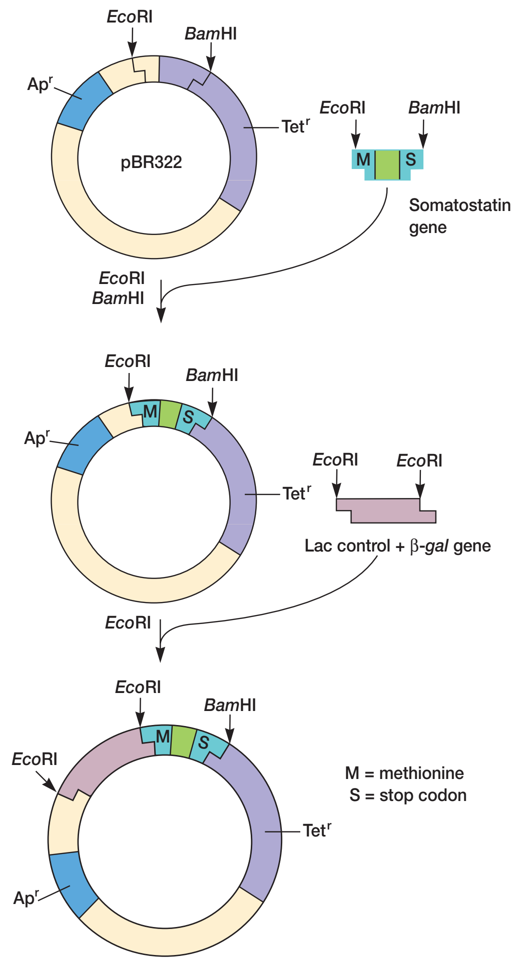


Figure 14.18 Cloning the Somatostatin Gene. An overview of the procedure used to synthesize a recombinant plasmid containing the somatostatin gene. See text for details.

After introduction of this chimeric plasmid into *E. coli,* the somatostatin gene was transcribed with the β-galactosidase gene fragment to generate an mRNA having both messages. Translation formed a protein consisting of the total hormone polypeptide attached to the β-galactosidase fragment by a methionine residue. Cyanogen bromide cleaves peptide bonds at methionine residues. Treatment of the fusion protein with cyanogen bromide broke the peptide chain at the methionine and released the hormone (**figure 14.19**). Once free, the polypeptide was able to fold properly and become active. Since production of the fusion protein was under the control of the lac operon, it could be easily regulated.

Many proteins have been produced since the synthesis of somatostatin. A similar approach was used to manufacture human insulin. Human growth hormone and some interferons also have been synthesized by rDNA techniques. The human growth hormone gene was too long to synthesize by chemical procedures and was prepared from mRNA as cDNA.

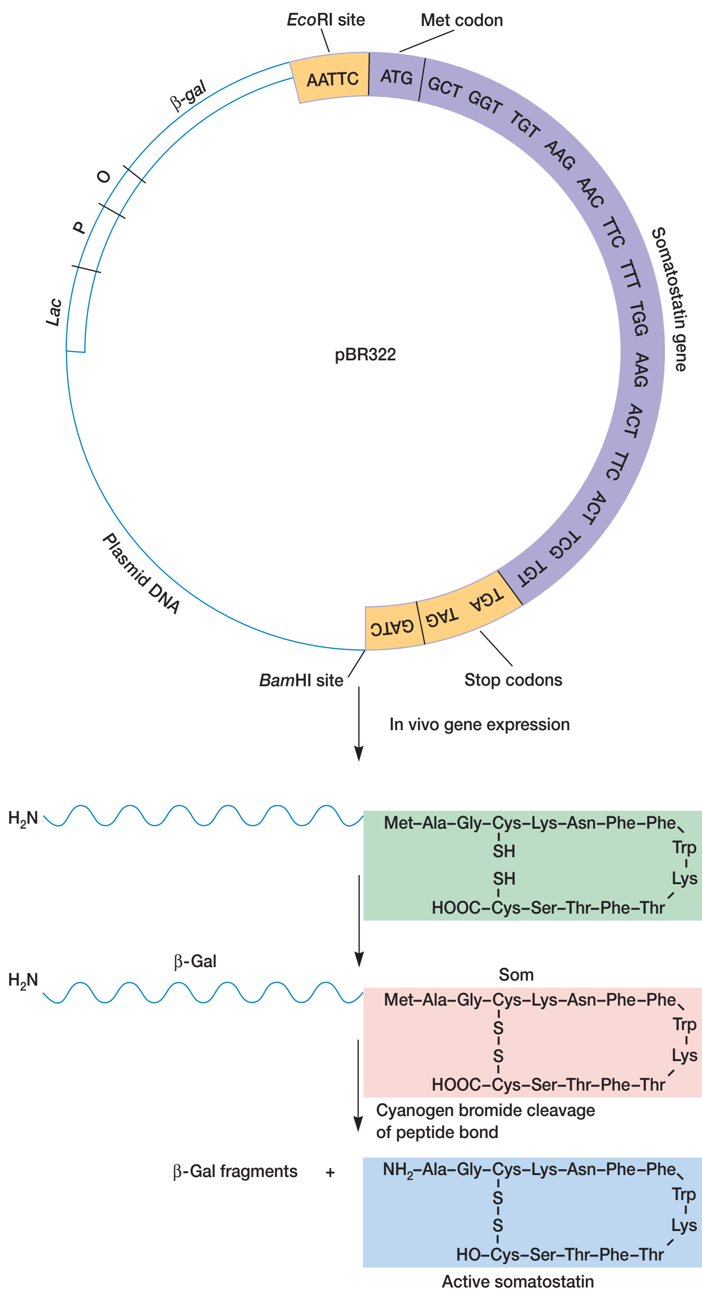


Figure 14.19 The Synthesis of Somatostatin by Recombinant *E. coli.* Cyanogen bromide cleavage at the methionine residue releases active hormone from the β-galactosidase fragment. The gene and associated sequences are shaded in color. Stop codons, the special methionine codon, and restriction enzyme sites are enclosed in boxes.

As mentioned earlier, introns in eucaryotic genes are not removed by bacteria and will render the final protein nonfunctional. The easiest solution is to prepare cDNA from processed mRNA that lacks introns and directly reflects the correct amino acid sequence of the protein product. In this instance it is particularly important to fuse the gene with an expression vector since a promoter and other essential sequences will be missing in the cDNA. Eucaryotic mRNA processing (pp. 263–64)

If the mRNA is scarce, it may not be easy to obtain enough for cDNA synthesis. Often the sequence of the protein coded for by the gene is used to deduce the best DNA sequence for the specific polypeptide segment (a logical procedure called reverse translation). Then the DNA probe is synthesized and used to locate and isolate the desired mRNA after gel electrophoresis. Finally, the isolated mRNA is used to make cDNA.

1. What is a transgenic animal? Describe how electroporation and gene guns are used to insert foreign genes into eucaryotic cells. What other approaches may be used? How are bacteria transformed?
2. How can one prevent rDNA from undergoing recombination in a bacterial host cell?
3. List several reasons why a cloned gene might not be expressed in a host cell. What is an expression vector?
4. Briefly outline the procedure for somatostatin production.
5. Identify a way to eliminate eucaryotic introns during the synthesis of rDNA.

14.8 Applications of Genetic Engineering

Genetic engineering and biotechnology will continue to contribute in the future to medicine, industry, and agriculture, as well as to basic research (**Box 14.1**). In this section some practical applications are briefly discussed.

Medical Applications

Certainly the production of medically useful proteins such as somatostatin, insulin, human growth hormone, and interferon is of great practical importance (**table 14.4**). This is particularly true of substances that previously only could be obtained from human tissues. For example, in the past, human growth hormone for treatment of pituitary dwarfism was extracted from pituitaries obtained during autopsies and was available only in limited amounts. Interleukin-2 (a protein that helps regulate the immune response) and blood-clotting factor VIII have recently been cloned, and undoubtedly other important peptides and proteins will be produced in the future. A particularly interesting development is the use of transgenic corn and soybean plants to produce monoclonal antibodies (*see p. 743*) for medical uses. It also is possible to use genetically engineered plants to produce oral vaccines. Genetically engineered mice now can produce fully human monoclonal antibodies. Synthetic vaccines—for instance, vaccines for malaria and rabies—are also being developed with recombinant techniques. A recombinant hepatitis B vaccine is already commercially available.

Box 14.1

Gene Expression and Kittyboo Colors

Genetic engineering techniques can yield unusual approaches to long-standing problems. To understand how the regulation of gene expression influences complex processes, the activity of more than one gene must be followed simultaneously. The use of genetic engineering techniques with luciferase genes has made this much easier.

A Jamaican beetle named the kittyboo has two light organs on its head and one on its abdomen. Recently four luciferase genes have been isolated from the beetle. Each luciferase enzyme produces a different colored light when it acts on the substrate luciferin. Keith V. Wood has cloned the genes and inserted them into *E. coli.* When the bacteria are exposed to luciferin, they glow (see **Box figure**).

These luciferase genes can be used to study gene regulation. Suppose that one wishes to follow the activity of the liver gene that codes for serum albumin. The albumin gene can be replaced with a luciferase gene, and the modified cell incubated with luciferin. Whenever the albumin gene is activated, the newly inserted luciferase gene will function and the cell glow. Measurement of light intensity is easy, rapid, and sensitive. By substituting a different luciferase gene for another liver gene, one can simultaneously follow the activity and coordination of two different liver genes. It is only necessary to measure light intensity at the two wavelengths characteristic of the luciferase genes.

***E. coli* with Luciferase Genes.** These four streaks of *E. coli* glow different colors because they contain four different luciferase genes cloned from the Jamaican click beetle or kittyboo, *Pyrophorus plagiophthalamus.*

Table 14.4 Some Human Peptides and Proteins Synthesized by Genetic Engineering

Peptide or Protein	Potential Use
α_1-antitrypsin	Treatment of emphysema
α-, β-, and γ-interferons	As antiviral, antitumor, and anti-inflammatory agents
Blood-clotting factor VIII	Treatment of hemophilia
Calcitonin	Treatment of osteomalacia
Epidermal growth factor	Treatment of wounds
Erythropoetin	Treatment of anemia
Growth hormone	Growth promotion
Insulin	Treatment of diabetes
Interleukins-1, 2, and 3	Treatment of immune disorders and tumors
Macrophage colony stimulating factor	Cancer treatment
Relaxin	Aid to childbirth
Serum albumin	Plasma supplement
Somatostatin	Treatment of acromegaly
Streptokinase	Anticoagulant
Tissue plasminogen activator	Anticoagulant
Tumor necrosis factor	Cancer treatment

Other medical uses of genetic engineering are being investigated. Probes are now being used in the diagnosis of infectious disease. An individual could be screened for mutant genes with probes and hybridization techniques (even before birth when used together with amniocentesis). A type of genetic surgery called somatic cell gene therapy may be possible for afflicted individuals. For example, cells of an individual with a genetic disease could be removed, cultured, and transformed with cloned DNA containing a normal copy of the defective gene or genes. These cells could then be reintroduced into the individual; if they became established, the expression of the normal genes might cure the patient. An immune deficiency disease patient lacking the enzyme adenosine deaminase that destroys toxic metabolic by-products has been treated in this way. Some of the patient's lymphocytes (*see p. 705*) were removed, given the adenosine deaminase gene with the use of a modified retrovirus, and returned to the patient's body. It may be possible to use a defective retrovirus (*see section 18.2*) or another virus to directly insert the proper genes into host cells, perhaps even specifically targeted organs or tissues. Fusion toxins provide a third example of potentially important genetic engineering applications. The first to be developed are recombinant proteins in which the enzymatic and membrane translocation domains of the diphtheria toxin (*see pp. 797–98*) are combined with proteins that attach specifically to target cell surface receptors. Clinical trials are being carried out on a fusion toxin that contains interleukin-2. IL-2 binds to the cell, and the diphtheria toxin enzymatic domain enters and blocks protein synthesis. Encouraging results have been obtained in the treatment of leukemia, lymphomas, and rheumatoid arthritis. The use of probes and plasmid fingerprinting in clinical microbiology (pp. 843–41)

It now appears that livestock will also be important in medically oriented biotechnology through the use of an approach sometimes called molecular pharming. Pig embryos injected with human hemoglobin genes develop into transgenic pigs that synthesize human hemoglobin. Current plans are to purify the hemoglobin and use it as a blood substitute. A pig could yield 20 units of blood substitute a year. Somewhat similar techniques have produced transgenic goats whose milk contains up to 3 grams of human tissue plasminogen activator per liter. Tissue plasminogen activator (TPA) dissolves blood clots and is used to treat cardiac patients.

Recombinant DNA techniques are playing an increasingly important role in research on the molecular basis of disease. The transfer from research to practical application often is slow because it is not easy to treat a disease effectively unless its molecular mechanism is known. Thus genetic engineering may aid in the fight against a disease by providing new information about its nature, as well as by aiding in diagnosis and therapy.

Industrial Applications

Industrial applications of recombinant DNA technology include manufacturing protein products by using bacteria, fungi, and cultured mammalian cells as factories; strain improvement for existing bioprocesses; and the development of new strains for additional bioprocesses. As mentioned earlier, the pharmaceutical industry is already producing several medically important polypeptides using this technology. In addition, there is interest in making expensive, industrially important enzymes with recombinant bacteria. Bacteria that metabolize petroleum and other toxic materials have been developed. These bacteria can be constructed by assembling the necessary catabolic genes on a single plasmid and then transforming the appropriate organism. There also are many potential applications in the chemical and food industries. Food microbiology (chapter 41); Industrial microbiology and biotechnology (chapter 42)

Agricultural Applications

It is also possible to bypass the traditional methods of selective breeding and directly transfer desirable traits to agriculturally important animals and plants. Potential exists for increasing growth rates and overall protein yields of farm animals. The growth hormone gene has already been transferred from rats to mice, and both in vitro fertilization and embryo implantation methods are fairly well developed. Recently, recombinant bovine growth hormone has been used to increase milk production by at least 10%. Perhaps farm animals' disease resistance and tolerance to environmental extremes also can be improved.

Cloned genes can be inserted into plant as well as animal cells. Presently a popular way to insert genes into plants is with a recombinant **Ti plasmid** (tumor-inducing plasmid) obtained from the bacterium *Agrobacterium tumefaciens* (**Box 14.2;** *see also section 30.4*). It also is possible to donate genes by forming plant cell protoplasts, making them permeable to DNA, and then adding the desired rDNA. The advent of the gene gun (p. 335) will greatly aid the production of transgenic plants.

Much effort has been devoted to the transfer of the nitrogen-fixing abilities of bacteria associated with legumes to other crop plants. The genes largely responsible for the process have been cloned and transferred to the genome of plant cells; however, the recipient cells have not been able to fix nitrogen. If successful, the potential benefit for crop plants such as corn is great. Nevertheless, there is some concern that new nitrogen-fixing varieties might spread indiscriminantly like weeds or disturb the soil nitrogen cycle (*see sections 28.3 and 30.4*).

Attempts at making plants resistant to environmental stresses have been more successful. For example, the genes for detoxification of glyphosate herbicides were isolated from *Salmonella,* cloned, and introduced into tobacco cells using the Ti plasmid. Plants regenerated from the recombinant cells were resistant to the herbicide. Herbicide-resistant varieties of cotton and fertile, transgenic corn also have been developed. This is of considerable importance because many crop plants suffer stress when treated with herbicides. Resistant crop plants would not be stressed by the chemicals being used to control weeds, and yields would presumably be much greater.

U.S. farmers grow substantial amounts of genetically modified (GM) crops. About a third of the corn, half of the soybeans, and a significant fraction of cotton crops are genetically modified. Cotton and corn are resistant to herbicides and insects. Soybeans have herbicide resistance and lowered saturated fat content. Other examples of genetically engineered commercial crops are canola, potato, squash, and tomato.

Many new agricultural applications are being explored. A good example is an altered strain of *Pseudomonas syringae* that protects against plant frost damage because it cannot produce the protein that induces ice-crystal formation. Much effort is being devoted to defending plants against pests without the use of chemical pesticides. A strain of *Pseudomonas fluorescens* carrying the gene for the *Bacillus thuringiensis* toxin (*see pp. 1020–21*) is under testing. This toxin destroys many insect pests such as the cabbage looper and the European corn borer. A variety of corn with the *B. thuringiensis* toxin gene has been developed. There is considerable interest in insect-killing viruses and particularly in the baculoviruses. A scorpion toxin gene has been inserted into the autographa californica multicapsid nuclear polyhedrosis virus (AcMNPV). The engineered AcMNPV kills cabbage looper more rapidly than the normal virus and reduces crop damage significantly. Finally, virus-resistant strains of soybeans, potatoes, squash, rice, and other plants are under development.

1. List several important present or future applications of genetic engineering in medicine, industry, and agriculture.
2. What is the Ti plasmid and why is it so important?

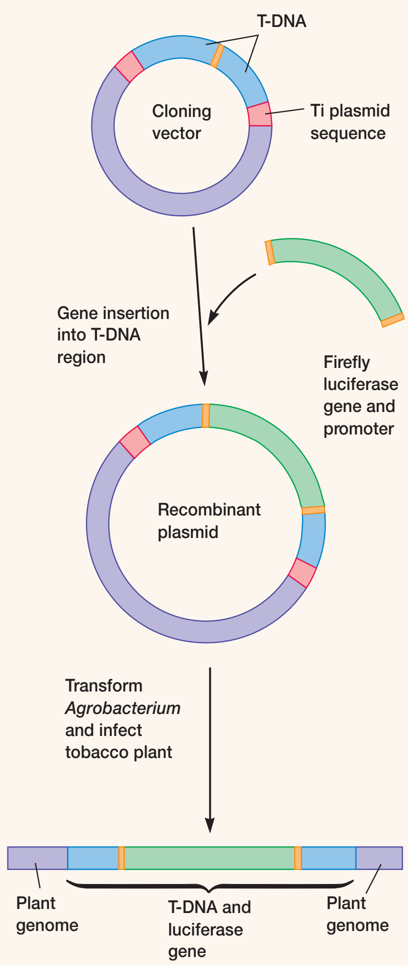
Box 14.2

Plant Tumors and Nature's Genetic Engineer

A plasmid from the plant pathogenic bacterium *Agrobacterium tumefaciens* is responsible for much success in the genetic engineering of plants. Infection of normal cells by the bacterium transforms them into tumor cells, and the crown gall disease develops in dicotyledonous plants such as grapes and ornamental plants. Normally, the gall or tumor is located near the junction of the plant's root and stem. The tumor forms because of the insertion of genes into the plant cell genome, and only strains of *A. tumefaciens* possessing a large conjugative plasmid called the Ti plasmid are pathogenic (*see Figure 30.20*). The Ti plasmid carries genes for virulence and the synthesis of substances involved in the regulation of plant growth. The genes that induce tumor formation reside between two 23 base pair direct-repeat sequences. This region is known as T-DNA and is very similar to a transposon. T-DNA contains genes for the synthesis of plant growth hormones (an auxin and a cytokinin) and an amino acid derivative called opine that serves as a nutrient source for the invading bacteria. In diseased plant cells, T-DNA is inserted into the chromosomes at various sites and is stably maintained in the cell nucleus.

When the molecular nature of crown gall disease was recognized, it became clear that the Ti plasmid and its T-DNA had great potential as a vector for the insertion of rDNA into plant chromosomes. In one early experiment the yeast alcohol dehydrogenase gene was added to the T-DNA region of the Ti plasmid. Subsequent infection of cultured plant cells resulted in the transfer of the yeast gene. Since then, many modifications of the Ti plasmid have been made to improve its characteristics as a vector. Usually one or more antibiotic resistance genes are added, and the nonessential T-DNA, including the tumor inducing genes, is deleted. Those genes required for the actual infection of the plant cell by the plasmid are left. T-DNA also has been inserted into the *E. coli* pBR322 plasmid and other plasmids to produce cloning vectors that can move between bacteria and plants (see **Box figure**). The gene or genes of interest are spliced into the T-DNA region between the direct repeats. Then the plasmid is returned to *A. tumefaciens,* plant culture cells are infected with the bacterium, and transformants are selected by screening for antibiotic resistance (or another trait in the T-DNA). Finally, whole plants are regenerated from the transformed cells. In this way several plants have been made herbicide resistant (p. 339).

Unfortunately *A. tumefaciens* does not produce crown gall disease in monocotyledonous plants such as corn, wheat, and other grains; and it has been used only to modify plants such as potato, tomato, celery, lettuce, and alfalfa. However, there is evidence that T-DNA is transferred to monocotyledonous plants and expressed, although it does not produce tumors. This discovery plus the creation of new procedures for inserting DNA into plant cells may well lead to the use of rDNA techniques with many important crop plants.



(a)

(b)

The Use of the Ti Plasmid Vector to Produce Transgenic Plants. (**a**) The formation of a cloning vector and its use in transformation. See text for details. (**b**) A tobacco plant, *Nicotiana tabacum,* that has been made bioluminescent by transfection with a special Ti plasmid vector containing the firefly luciferase gene. When the plant is watered with a solution of luciferin, the substrate for the luciferase enzyme, it glows. The picture was made by exposing the transgenic plant to Ektachrome film for 24 hours.

14.9 Social Impact of Recombinant DNA Technology

Despite the positive social impact of rDNA technology, dangers may be associated with rDNA work and gene cloning. The potential to alter an organism genetically raises serious scientific and philosophical questions, many of which have not yet been adequately addressed. In this section some of the debate is briefly reviewed.

The initial concern raised by the scientific community was that recombinant *E. coli* and other genetically engineered microorganisms (GEMs) carrying dangerous genes might escape and cause widespread infections. Because of these worries, the federal government established guidelines to limit and regulate the locations and types of potentially dangerous experiments. Physical containment was to be practiced—that is, rDNA research was to be carried out in specially designed laboratories with extra safety precautions. In addition, rDNA researchers were also to practice biological containment. Only weakened microbial hosts unable to survive in a natural environment and nonconjugating bacterial strains were to be used to avoid the spread of the vector or rDNA itself. Further experiments suggested that the dangers were not as extreme as initially conceived. Yet little is known about what would happen if recombinant organisms escaped. For example, could dangerous genes such as oncogenes (*see section 18.5*) move from a weakened strain to a hardy one that would then spread? Will there be increased risk when extremely large quantities of recombinant microorganisms are grown in industrial fermenters? Some people worry because the use of rDNA techniques has become so widespread and the guidelines on their safe use have been relaxed to a considerable extent. Biomedical rDNA research has been regulated by the Recombinant DNA Advisory Committee (RAC) of the National Institutes of Health. More recently, the Food and Drug Administration (FDA) has assumed principal responsibility for oversight of gene therapy research. The Environmental Protection Agency and state governments have jurisdiction over agriculturally related field experiments. Industrial rDNA research can proceed without such close regulation, and this has also caused some anxiety about potential safety hazards.

Aside from these concerns over safety, the use of recombinant technology on human beings raises ethical and moral questions. These problems are not extreme in cases of somatic cell gene therapy to cure serious disease. There is much greater concern about attempts to correct defects or bestow traits considered to be desirable or cosmetic by introducing genes into human eggs and embryos. In 1983 two Nobel Prize winners, several other prominent scientists, and about 40 religious leaders signed a statement to Congress requesting that such alterations of human eggs or sperm not be attempted. There is a temptation to "improve" ourselves or reengineer the human body. Some might wish to try to change their children's intelligence, size, or physical attractiveness through genetic modifications. Others argue that the good arising from such interventions more than justifies the risks of misuse and that there is nothing inherently wrong in modifying the human gene pool to reduce the incidence of genetic disease. This discussion extends to other organisms as well. Some argue that we have no right to create new life-forms and further disrupt the genetic diversity that human activities have already severely reduced. United States courts have decided that researchers and companies can patent "nonnaturally occurring" living organisms, whether microbial, plant, or animal.

One of the major current efforts in biotechnology, the human genome project, has determined the sequences of all human chromosomes. Success in this project and further technical advances in biotechnology will make genetic screening very effective. Physicians will be able to detect critical flaws in DNA long before the resulting genetic disease becomes manifest. In some cases this may lead to immediate treatment. But often nothing can be done about the damage and all sorts of dilemmas arise. Should people be told about the situation? Do they even wish to know about defects that cannot be corrected? What happens if their employers and insurance companies learn of the problem? Employees may lose their jobs and insurance. The issue of privacy is crucial in a practical sense, especially if genetic data banks are established. There may be increasing pressure on parents to abort fetuses with genetic defects in order not to stress the medical and social welfare systems. Clearly modern biomedical technology holds both threat and promise. Society has not yet faced its implications despite rapid progress in the development of genetic screening technology.

Another area of considerable controversy involves the use of recombinant organisms in agriculture. Many ecologists are worried that the release of unusual, recombinant organisms without careful prior risk assessment may severely disrupt the ecosystem. They point to the many examples of ecosystem disruption from the introduction of foreign organisms. A major concern is the exchange of genes between transgenic plants and viruses or other plants. Viral nucleic acids inserted into plants to make them virus resistant might combine with the genome of an invading virus to make it even more virulent. Weeds might acquire resistance to viruses, pests, and herbicides from transgenic crop plants. Other potential problems trouble critics. For example, insect-killing viruses might destroy many more insect species than expected. Toxin genes inserted into one virus might move to other viruses. Genetically modified foods might trigger damaging allergic responses in consumers. Potential risks have been vigorously discussed because several recombinant organisms are already on the market. Thus far, no obvious ecological or health effects have been observed. Nevertheless, there have been calls for stricter regulation of GM foods. In Europe, GM crops are not commonly produced by farmers or purchased by consumers. Some large U.S. food producers have responded to public concern and quit using GM crops.

As with any technology, the potential for abuse exists. A case in point is the use of genetic engineering in biological warfare and terrorism. Although there are international agreements that limit the research in this area to defense against other biological weapons, the knowledge obtained in such research can easily be used in offensive biological warfare. Effective vaccines constructed using rDNA technology can protect the attacker's troops and civilian population. Because it is relatively easy and inexpensive to prepare bacteria capable of producing massive quantities of toxins or to develop particularly virulent strains of viral and bacterial pathogens, even small countries and terrorist organizations might acquire biological weapons. Many scientists and nonscientists also are concerned about increases in the military rDNA research carried out by the major powers. The emerging threat of bioterrorism (p. 863)

Recombinant DNA technology has greatly enhanced our knowledge of genes and how they function, and it promises to improve our lives in many ways. Yet, as this brief discussion shows, problems and concerns remain to be resolved. Past scientific advances have sometimes led to unanticipated and unfortunate consequences, such as environmental pollution and nuclear weapons. With prudence and forethought we may be able to avoid past mistakes in the use of this new technology.

1. Describe four major areas of concern about the application of genetic engineering. In each case give both the arguments for and against the use of genetic engineering.

Summary

1. Genetic engineering became possible after the discovery of restriction enzymes and reverse transcriptase, and the development of both the Southern blotting technique and other essential methods in nucleic acid chemistry (**table 14.1**).
2. Oligonucleotides of any desired sequence can be synthesized by a DNA synthesizer machine. This has made possible site-directed mutagenesis.
3. The polymerase chain reaction allows small amounts of DNA to be increased in concentration thousands of times (**figure 14.8**).
4. Agarose gel electrophoresis is used to separate DNA fragments according to size differences.
5. Fragments can be isolated and identified, then joined with plasmids or phage genomes and cloned; one also can first clone all fragments and subsequently locate the desired clone. (**figures 14.11, 14.14, and 14.15**).
6. Three techniques that can be used to join DNA fragments are the creation of similar sticky ends with a single restriction enzyme, the addition of poly(dA) and poly(dT) to create sticky ends, and blunt-end ligation with T4 DNA ligase.
7. Probes for the detection of recombinant clones are made in several ways and usually labeled with ^{32}P. Nonradioactive probes are also used.
8. Several types of vectors may be used, each with different advantages: plasmids, phages and other viruses, cosmids, artificial chromosomes, and shuttle vectors (**table 14.3**).
9. Genes can be inserted into eucaryotic cells by techniques such as microinjection, electroporation, and the use of a gene gun.
10. The recombinant vector often must be modified by the addition of promoters, leaders, and other elements. Eucaryotic gene introns also must be removed. An expression vector has the necessary features to express any recombinant gene it carries.
11. Many useful products, such as the hormone somatostatin, have been synthesized using recombinant DNA technology (**figure 14.19**).
12. Recombinant DNA technology will provide many benefits in medicine, industry, and agriculture.
13. Despite the great promise of genetic engineering, it also brings with it potential problems in areas of safety, human experimentation, potential ecological disruption, and biological warfare.

Key Terms

autoradiography *322*
bacterial artificial chromosome (BAC) *335*
biotechnology *320*
chimera *334*
complementary DNA (cDNA) *321*
cosmid *335*
electrophoresis *327*
electroporation *335*
expression vector *336*
gene gun *335*
genetic engineering *320*
library *330*
oligonucleotides *323*
polymerase chain reaction (PCR technique) *326*
probe *322*
recombinant DNA technology *320*
restriction enzymes *320*
site-directed mutagenesis *323*
Southern blotting technique *322*
Ti plasmid *339*
transfection *335*
transgenic animal *335*
vectors *322*
yeast artificial chromosome (YAC) *335*

Questions for Thought and Review

1. Could the Southern blotting technique be applied to RNA and proteins? How might this be done?
2. Why could a band on an electrophoresis gel still contain more than one kind of DNA fragment?
3. What advantage might there be in creating a genomic library first rather than directly isolating the desired DNA fragment?
4. In what areas do you think genetic engineering will have the greatest positive impact in the future? Why?
5. What do you consider to be the greatest potential dangers of genetic engineering? Are there ethical problems with any of its potential applications?

Critical Thinking Questions

1. Initial attempts to perform PCR were carried out using the DNA polymerase from *E. coli.* What was the major difficulty?
2. Why can't PCR be done with RNA polymerase?
3. Write a one-page explanation for a reader with a high school education that explains gene therapy. Assume the reader has a son or daughter with a life-threatening disease for which the only treatment course is gene therapy.
4. Suppose that one inserted a simple plasmid (one containing an antibiotic resistance gene and a separate restriction site) carrying a human interferon gene into *E. coli,* but none of the transformed bacteria produced interferon. Give as many plausible reasons as possible for this result.

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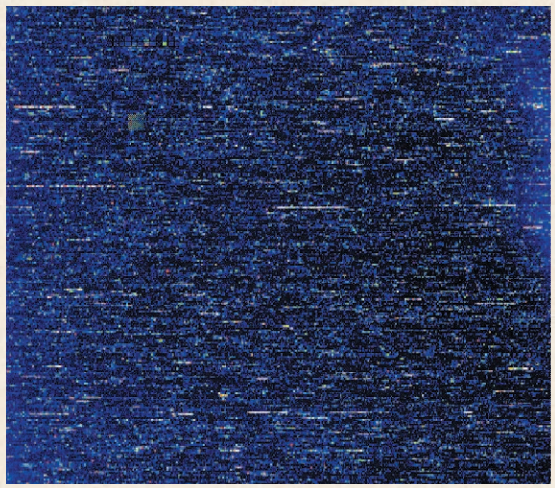
CHAPTER 15

Microbial Genomics



A DNA chip can be used to follow gene expression by a microorganism's complete genome. This chip contains probes for the over 4,200 known open reading frames in the *E. coli* genome.

Outline

Concepts

1. Genomics is the study of the molecular organization of genomes, their information content, and the gene products they encode. It may be divided into structural genomics, functional genomics, and comparative genomics.
2. Individual pieces of DNA can be sequenced using the Sanger method. The easiest way to analyze microbial genomes is by whole-genome shotgun sequencing in which randomly produced fragments are sequenced individually and then aligned by computers to give the complete genome.
3. Because of the mass of data to be analyzed, the use of sophisticated programs on high-speed computers is essential to genomics.
4. Many bacterial genomes have already been sequenced and compared. The results are telling us much about such subjects as genome structure, microbial physiology, microbial phylogeny, and how pathogens cause disease. They will undoubtedly help in preparing new vaccines and drugs for the treatment of infectious disease.
5. Genome function can be analyzed by annotation, the use of DNA chips to study mRNA synthesis, and the study of the organism's protein content (proteome) and its changes.

> *A prerequisite to understanding the complete biology of an organism is the determination of its entire genome sequence.*
>
> —*J. Craig Venter, et al.*

Chapter 13 provided a brief introduction to microbial recombination and plasmids, including the use of conjugation and other techniques in mapping the chromosome. Chapter 14 described the development and impact of recombinant DNA technology. This chapter will carry these themes further with the discussion of the current revolution in genome sequencing. We will begin with a general overview of the topic, followed by an introduction to the DNA sequencing technique. Next, the whole-genome shotgun sequencing method will be briefly described. This is followed by a comparison of selected microbial genomes and a discussion of what has been learned from them. After we have considered genome structure, we will turn to genome function and the array of transcripts and proteins produced by genomes. The focus will be on annotation, DNA chips, and the use of two-dimensional electrophoresis to study the proteome. The chapter concludes with a brief consideration of future challenges and opportunities in genomics.

15.1 Introduction

Genomics is the study of the molecular organization of genomes, their information content, and the gene products they encode. It is a broad discipline, which may be divided into at least three general areas. **Structural genomics** is the study of the physical nature of genomes. Its primary goal is to determine and analyze the DNA sequence of the genome. **Functional genomics** is concerned with the way in which the genome functions. That is, it examines the transcripts produced by the genome and the array of proteins they encode. The third area of study is **comparative genomics,** in which genomes from different organisms are compared to look for significant differences and similarities. This helps identify important, conserved portions of the genome and discern patterns in function and regulation. The data also provide much information about microbial evolution, particularly with respect to phenomena such as horizontal gene transfer.

It should be emphasized at the beginning that whole-genome sequence information provides an entirely new starting point for biological research. In the future, microbiologists will not have to spend as much time cloning genes because they will be able to generate new questions and hypotheses from computer analyses of genome data. Then they can test their hypotheses in the laboratory.

15.2 Determining DNA Sequences

The most widely used sequencing technique is that developed by Frederick Sanger in 1975. This approach uses dideoxynucleoside triphosphates (ddNTPs) in DNA synthesis. These molecules resemble normal nucleotides except that they lack a 3′-hydroxyl group (**figure 15.1**). They are added to the growing end of the chain, but terminate the synthesis catalyzed by DNA polymerase because more nucleotides cannot be attached to further extend the chain. In the manual sequencing method, a single strand of the DNA to be sequenced is mixed with a primer, DNA polymerase I, four deoxynucleoside triphosphate substrates (one of which is radiolabeled), and a small amount of one of the dideoxynucleotides. DNA synthesis begins with the primer and terminates when a ddNTP is incorporated in place of a regular deoxynucleoside triphosphate. The result is a series of fragments of varying lengths. Four reactions are run, each with a different ddNTP. The mix with ddATP produces fragments with an A terminus; the mix with ddCTP produces fragments with C terminals, and so forth (**figure 15.2**). The radioactive fragments are removed from the DNA template and electrophoresed on a polyacrylamide gel to separate them from one another based on size. Four lanes are electrophoresed, one for each reaction mix, and the gel is autoradiographed (*see p. 322*). A DNA sequence is read directly from the gel, beginning with the smallest fragment or fastest-moving band and moving to the largest fragment or slowest band (figure 15.2*a*). Up to 800 residues can be read from a single gel.

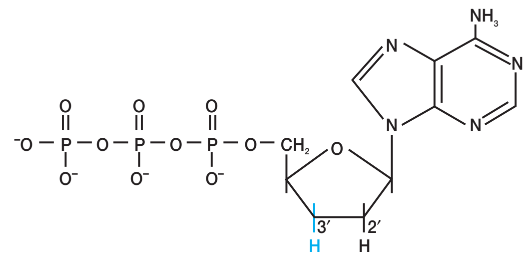


Figure 15.1 Dideoxyadenosine triphosphate (ddATP). Note the lack of a hydroxyl group on the 3′ carbon, which prevents further chain elongation by DNA polymerase.

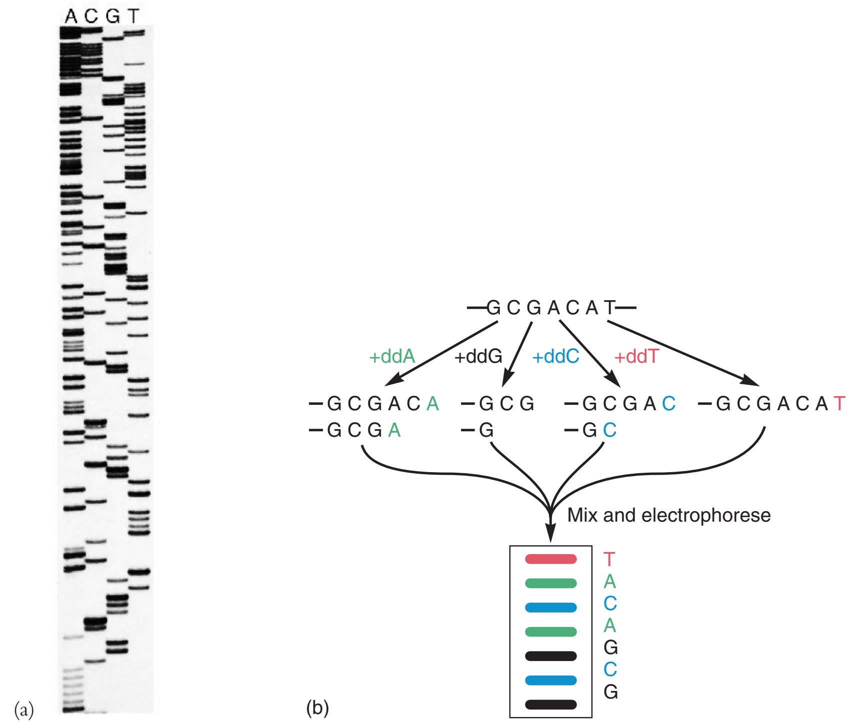
In automated systems dideoxynucleotides that have been labeled with fluorescent dyes are used (each ddNTP is labeled with a dye of a different color). The products from the four reactions are mixed and electrophoresed together. Because each ddNTP fluoresces with a different color, a detector can scan the gel and rapidly determine the sequence from the order of colors in the bands (figure 15.2*b,c*).

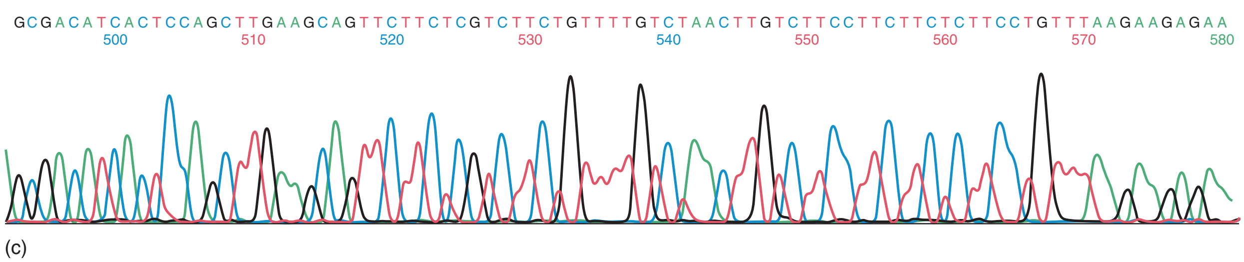
Recently, fully automated capillary electrophoresis sequencers have been developed. These are much faster and allow up to 96 samples to be sequenced simultaneously; it is possible to generate over 350 kilobases of sequences a day. Current systems can sequence strands of DNA around 700 bases long in about 4 hours.

15.3 Whole-Genome Shotgun Sequencing

Although several virus genomes have been sequenced, in the past it has not been possible to sequence the genomes of bacteria. Prior to 1995, whole-genome approaches to sequencing were not

Figure 15.2 The Sanger Method for DNA Sequencing. (**a**) A sequencing gel with four separate lanes. The sequence begins, reading from the bottom, CAAAAAACGGACCGGGTGTAC. (**b**) An example of sequencing by use of fluorescent dideoxynucleoside triphosphates. See text for details. (**c**) Part of an automated DNA sequencing run. Bases 493 to 499 were used as the example in (*b*).





possible because available computational power was insufficient for assembling a genome from thousands of DNA fragments. J. Craig Venter, Hamilton Smith, and their collaborators initially sequenced the genomes of two free-living bacteria, *Haemophilus influenzae* and *Mycoplasma genitalium.* The genome of *H. influenzae,* the first to be sequenced, contains about 1,743 genes in 1,830,137 base pairs and is much larger than a virus genome.

Venter and Smith developed an approach called **whole-genome shotgun sequencing.** The process is fairly complex when considered in detail, and there are many procedures to ensure the accuracy of the results, but the following summary gives a general idea of the approach originally employed by The Institute of Genomic Research (TIGR). For simplicity, this approach may be broken into four stages: library construction, random sequencing, fragment alignment and gap closure, and editing.

1. *Library construction.* The large bacterial chromosomes were randomly broken into fairly small fragments, about the size of a gene or less, using ultrasonic waves; the fragments were then purified (**figure 15.3**). These fragments were attached to plasmid vectors (*see pp. 334–35*), and plasmids with a single insert were isolated. Special *E. coli* strains lacking restriction enzymes were transformed with the plasmids to produce a library of the plasmid clones.
2. *Random sequencing.* After the clones were prepared and the DNA purified, thousands of bacterial DNA fragments were sequenced with automated sequencers, employing special dye-labeled primers. Thousands of templates were used, normally with universal primers that recognized the plasmid DNA sequences just next to the bacterial DNA insert. The nature of the process is such that almost all stretches of genome are sequenced several times, and this increases the accuracy of the final results.
3. *Fragment alignment and gap closure.* Using special computer programs, the sequenced DNA fragments were clustered and assembled into longer stretches of sequence by comparing nucleotide sequence overlaps between fragments. Two

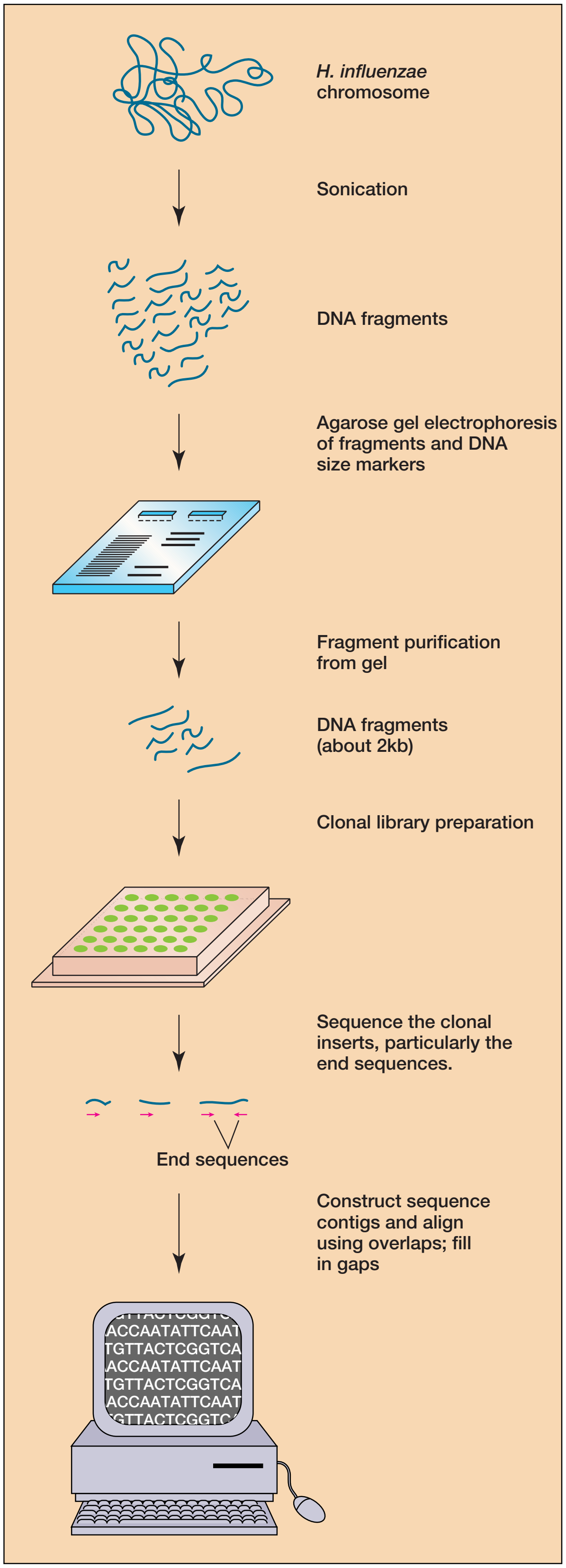


Figure 15.3 Whole-Genome Shotgun Sequencing. This general overview shows how the *Haemophilus influenzae* genome was sequenced. See text for details.

fragments were joined together to form a larger stretch of DNA if the sequences at their ends overlapped and matched (i.e., were the same). This overlap comparison process resulted in a set of larger contiguous nucleotide sequences or contigs.

Finally, the contigs were aligned in the proper order to form the completed genome sequence. If gaps existed between two contigs, sometimes fragment samples with their ends in the two adjacent contigs were available. These fragments could be analyzed and the gaps filled in with their sequences. When this approach was not possible, a variety of other techniques were used to align contigs and fill in gaps. For example, λ phage libraries containing large bacterial DNA fragments were constructed (*see pp. 330–332, 335*). The large fragments in these libraries overlapped the previously sequenced contigs. These fragments were then combined with oligonucleotide probes that matched the ends of the contigs to be aligned. If the probes bound to a λ library fragment, it could be used to prepare a stretch of DNA that represented the gap region. Overlaps in the sequence of this new fragment with two contigs would allow them to be placed side-by-side and fill in the gap between them.

4. *Editing.* The sequence was then carefully proofread in order to resolve any ambiguities in the sequence. Also the sequence was checked for unwanted frameshift mutations and corrected if necessary.

The approach worked so well that it took less than 4 months to sequence the *M. genitalium* genome (about 500,000 base pairs in size). The shotgun technique also has been used successfully by Celera Genomics in the Human Genome Project and to sequence the *Drosophila* genome.

Once the genome sequence has been established, the process of **annotation** begins. The goal of annotation is to determine the location of specific genes in the genome map. Every **open reading frame (ORF)**—a reading frame sequence (*see p. 241*) not interrupted by a stop codon—larger than 100 codons is considered to be a potential protein coding sequence. Computer programs are used to compare the sequence of the predicted ORF against large databases containing nucleotide and amino acid sequences of known enzymes and other proteins. If a bacterial sequence matches one in the database, it is assumed to code for the same protein. Although this comparison process is not without errors, it can provide tentative function assignments for about 40 to 50% of the presumed coding regions. It also gives some information about transposable elements, operons, repeat sequences, the presence of various metabolic pathways, and other genome features. The results of genome sequencing and annotation for *Mycoplasma genitalium* and *Haemophilus influenzae* are shown in figures 15.5 and 15.6. Often the results of annotation are expressed in a diagram that summarizes the known metabolism and physiology of the organism. An example of this is given in figure 15.7.

1. Define genomics. What are the three general areas into which it can be divided.
2. Describe the Sanger method for DNA sequencing.
3. Outline the whole-genome shotgun sequencing method. What is annotation and how is it carried out?

15.4 Bioinformatics

DNA sequencing techniques have developed so rapidly that an enormous amount of data has already accumulated and genomes are being sequenced at an ever-increasing pace. The only way to organize and analyze all these data is through the use of computers, and this has led to the development of a new interdisciplinary field that combines biology, mathematics, and computer science. **Bioinformatics** is the field concerned with the management and analysis of biological data using computers. In the context of genomics, it focuses on DNA and protein sequences. The annotation process just described is one aspect of bioinformatics. DNA sequence data is stored in large databases. One of the largest genome databases is the International Nucleic Acid Sequence Data Library, often referred to as GenBank. Databases can be searched with special computer programs to find homologous sequences, DNA sequences that are similar to the one being studied. Protein coding regions also can be translated into amino acid sequences and then compared. These sequence comparisons can suggest functions of the newly discovered genes and proteins. The gene under study often will have a function similar to that of genes with homologous DNA or amino acid sequences.

15.5 General Characteristics of Microbial Genomes

The development of shotgun sequencing and other genome sequencing techniques has led to the characterization of many procaryotic genomes in a very short time. Many genome sequences of procaryotes have been completed and published, and some of these are given in **table 15.1.** These procaryotes represent great phylogenetic diversity (**figure 15.4**). At least 100 more procaryotes, many of them major human pathogens, are being sequenced at present. Comparison of the genomes from different procaryotes will contribute significantly to the understanding of procaryotic evolution and help deduce which genes are responsible for various cellular processes. Genome sequences will aid in our understanding of genetic regulation and genome organization. In some cases, such information will also aid in the search for human genes by the Human Genome Project because of the similarities between procaryotic and human biochemistry.

Currently published genome sequences already have provided new and important insights into genome organization and function. *Mycoplasma genitalium* grows in human genital and respiratory tracts and has a genome of only 580 kilobases in size, one of the smallest genomes of any free-living organism (**figure 15.5**). Thus the sequence data are of great interest because they help establish the minimal set of genes needed for a free-living existence. There appear to be approximately 517 genes (480 protein-encoding genes and 37 genes for RNA species). About 90 proteins are involved in translation, and only around 29 proteins for DNA replication. Interestingly, 140 genes, or 29% of those in the genome, code for membrane proteins, and up to 4.5% of the genes seem to be involved in evasion of host immune responses. Only 5 genes have regulatory

Table 15.1 Examples of Complete Published Microbial Genomes

Genome	Domain[a]	Size (Mb)	% G + C
Aquifex aeolicus	B	1.50	43
Archaeoglobus fulgidus	A	2.18	48
Bacillus subtilis	B	4.20	43
Borrelia burgdorferi	B	1.44	28
Campylobacter jejuni	B	1.64	31
Chlamydia pneumoniae	B	1.23	40
Chlamydia trachomatis	B	1.05	41
Deinococcus radiodurans	B	3.28	67
Escherichia coli	B	4.60	50
Haemophilus influenzae Rd	B	1.83	39
Helicobacter pylori	B	1.66	39
Methanobacterium thermoautotrophicum	A	1.75	49
Methanococcus jannaschii	A	1.66	31
Mycobacterium tuberculosis	B	4.40	65
Mycoplasma genitalium	B	0.58	31
Mycoplasma pneumoniae	B	0.81	40
Neisseria meningitidis	B	2.27	51
Pseudomonas aeruginosa	B	6.3	67
Pyrococcus horiksohii	A	1.80	42
Rickettsia prowazekii	B	1.10	29
Saccharomyces cerevisiae	E	13	38
Synechocystis sp.	B	3.57	47
Thermotoga maritima	B	1.80	46
Treponema pallidum	B	1.14	52
Vibrio cholerae	B	4.0	48

[a]The following abbreviations are used: A, Archaea; B, Bacteria; E, Eucarya.

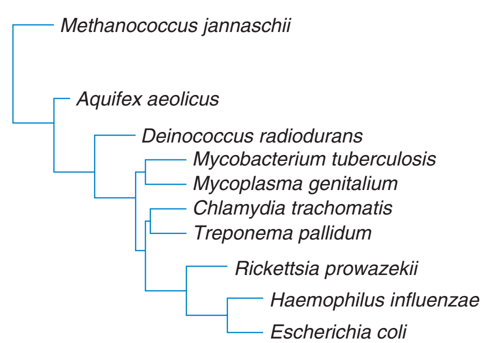


Figure 15.4 Phylogenetic Relationships of Some Procaryotes with Sequenced Genomes. These procaryotes are discussed in the text. *Methanococcus jannaschii* is in the domain *Archaea,* the rest are members of the domain *Bacteria.* Genomes from a broad diversity of procaryotes have been sequenced and compared. *Source: The Ribosomal Database Project.*

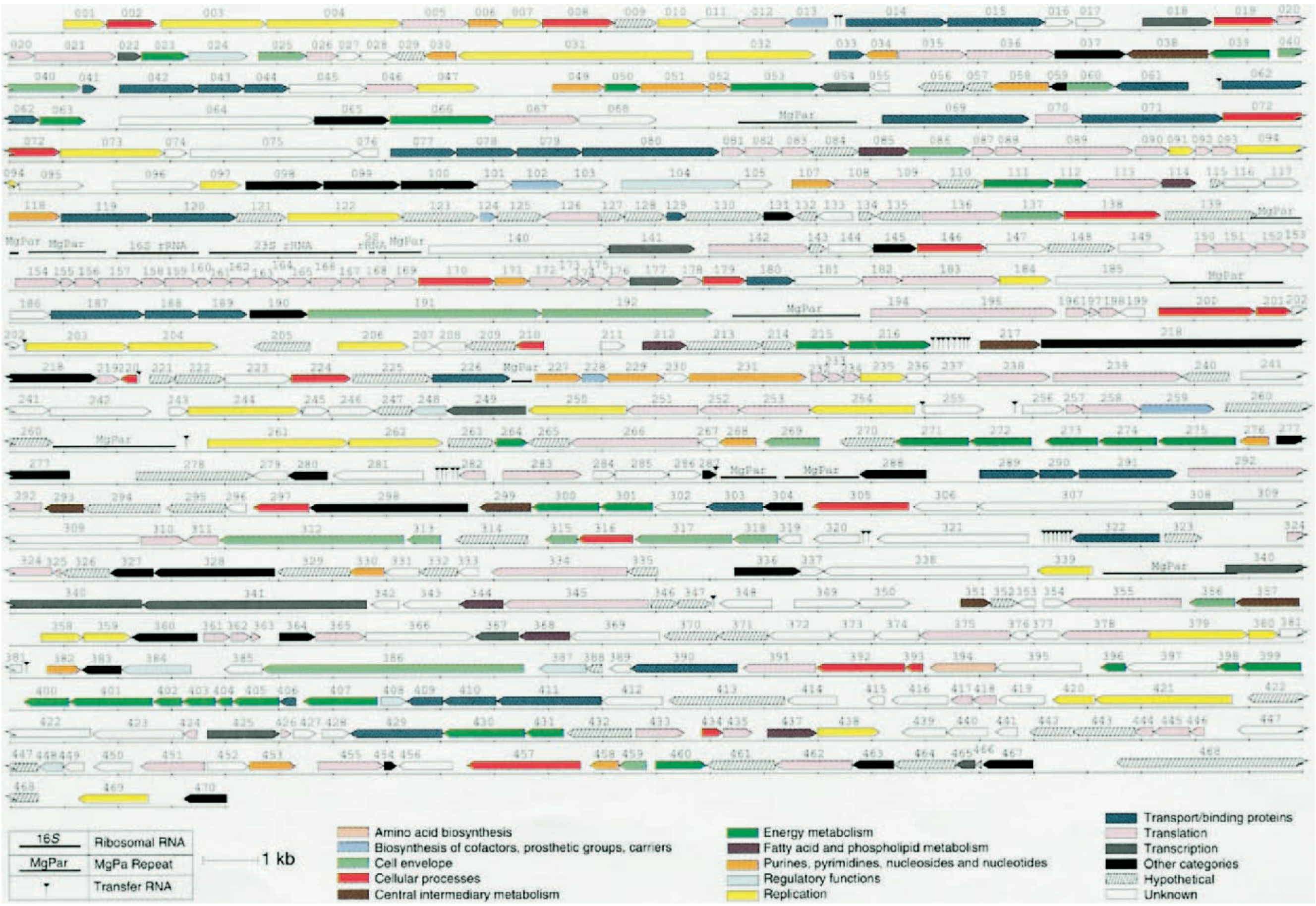


Figure 15.5 Map of the *Mycoplasma genitalium* genome. The predicted coding regions are shown with the direction of transcription indicated by arrows. The genes are color coded by their functional role. The rRNA operon, tRNA genes, and adhesin protein operons (MgPa) are indicated. *Reprinted with permission from Fraser, C. M., et al. Copyright 1995. The minimal gene complement of* Mycoplasma genitalium. Science *270:397–403. Figure 1, page 398 and The Institute for Genomic Research.*

functions. Even in this smallest genome, 22% of the genes do not match any known protein sequence. Comparison with the *M. pneumoniae* genome and studies of gene inactivation by transposon insertion suggests that about 108 to 121 *M. genitalium* genes may not be essential for survival. Thus the minimum gene set required for laboratory growth conditions seems to be approximately 265 to 350 genes; about 100 of these have unknown functions. *Haemophilus influenzae* has a much larger genome, 1.8 megabases and 1,743 genes (**figure 15.6**). More than 40% of the genes have unknown functions. It has already been found that the bacterium lacks three Krebs cycle genes and thus a functional cycle. It does devote many more genes (64 genes) to regulatory functions than does *M. genitalium. Haemophilus influenzae* is a species capable of transformation (*see pp. 305–7*). The process must be very important to this bacterium because it contains 1,465 copies of the recognition sequence used in DNA uptake during transformation. *Methanococcus jannaschii,* a member of the Archaea, also has been sequenced. Only 44% of its 1,738 genes match those of other organisms, an indication of how different this archaeon is from bacteria and eucaryotes. Despite this profound difference, many of its genes for DNA replication, transcription, and translation are similar to eucaryotic genes and quite different from bacterial genes. However, the metabolism of *M. jannaschii* is more similar to that of bacteria than to eucaryotic metabolism. More recently the sequence of the 4.6 megabase *Escherichia coli* K12 genome has been published. About 5 to 6% of the genes code for proteins involved in cell and membrane structure; 12 to 14% for transport proteins; 10% for the enzymes of energy and central intermediary metabolic pathways; 4% for regulatory genes; and 8% for replication, transcription, and translation proteins. The genome contains about 4,288 predicted genes, almost 2,500 of which do not resemble known genes. The large number of unknown genes in *Escherichia coli, Haemophilus influenzae,* and other procaryotes has great significance. It shows how little we know about microbial biology. Clearly there is much more to learn about the genetics, physiology, and metabolism of procaryotes, even of those that have been intensively studied.